

Compatibility-based Bayesian inference: asymptotic limits, variational principles, and Wasserstein geometry

Marco Tarsia · Antonietta Mira · Daniele Cassani

Abstract. Approximate Bayesian Computation (ABC) replaces intractable likelihood evaluations with comparisons between observed and simulated data. We develop a measure-theoretic formulation of discrepancy-based Bayesian inference in which compatibility is encoded through nonnegative weights. For vanishing thresholds, we establish convergence of the corresponding posterior distributions in total variation. In the large-sample regime, we identify conditions under which the posterior converges to the prior conditioned on an asymptotic compatibility region, showing that concentration need not occur at a fixed positive tolerance. We also derive persistence bounds under weaker assumptions and prove stability of the associated entropy-regularized variational problems. For Wasserstein discrepancies, we obtain a geometric representation through distances from compatibility balls and an exact variational reduction to probability distributions over model laws.

Key words. Approximate Bayesian Computation, compatibility filtering, entropy-regularized inference, optimal transport, Wasserstein distance.

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Introduction

Bayesian inference traditionally relies on the evaluation of a likelihood function relating the observed data to the underlying statistical model. In many applications, however, the data-generating mechanism can be simulated while the likelihood remains unavailable or computationally intractable. Approximate Bayesian Computation (ABC) replaces likelihood evaluation with comparisons between observed and simulated data through discrepancy measures; see, e.g., [3, 24, 37], as well as [12, 13, 7, 20]. The choice of discrepancy therefore plays a central role, influencing both the statistical behavior of posterior distributions and the structure of the inferential procedure. This has motivated the development of ABC methods based on distributional discrepancies, including Kullback–Leibler divergence, energy statistics, and kernel

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embeddings; see, e.g., [34, 21, 30]. Wasserstein distances have also attracted particular attention in statistics and machine learning, providing a geometric means of comparing probability distributions that reflects the underlying geometry of the observation space; see, e.g., [4, 32, 41, 23].

The asymptotic behavior of ABC posteriors, including concentration and model misspecification, has been extensively investigated; see, e.g., [16, 17]. A central question concerns the dependence of the posterior on the tolerance used to assess agreement between observed and simulated data. A smaller tolerance imposes a stricter compatibility requirement, but its implications for posterior concentration depend on the asymptotic regime. Two distinct limits arise naturally. One concerns the behavior of the posterior as the tolerance approaches zero while the sample size remains fixed. The other concerns increasing sample sizes, with the tolerance held fixed. These limits describe different aspects of the inferential procedure and need not produce the same posterior behavior. In particular, increasing the sample size does not necessarily force the posterior to concentrate around a single parameter value.

This distinction becomes clearer when inference is viewed in terms of compatibility between probability distributions. Different parameter values may generate sampling distributions that are sufficiently close to the observational distribution according to the chosen discrepancy. If the tolerance remains positive, several such values may continue to satisfy the compatibility criterion as the sample size increases. The resulting posterior may therefore retain a nontrivial distribution over a region of compatible parameters rather than concentrate on a single value. This possibility is particularly relevant under model misspecification, when the observational distribution does not belong to the assumed statistical model, or when different parameters generate indistinguishable or nearly indistinguishable sampling distributions.

A complementary direction concerns transport-based ABC, where discrepancies between empirical probability measures are expressed through Wasserstein distances; see [4]. A probabilistic interpretation of ABC acceptance rules in terms of model error is provided in [42]. Together, these contributions illustrate the importance of the probabilistic structure underlying discrepancy-based inference. A natural question is therefore how to characterize compatibility regions determined by the limiting observational distribution, the family of sampling distributions, and the chosen discrepancy. Such regions identify the parameter values that remain compatible with the observational law at a prescribed tolerance.

Generalized Bayesian inference allows posterior distributions to be constructed from a loss function and a prior distribution without requiring an explicit likelihood; see, e.g., [5]. Related approaches include Bayesian inference via coarsening, likelihood-free generalized Bayes based on scoring rules, and robust generalized Bayes based on Stein discrepancies; see, respectively, [28, 31, 26]. Compatibility weights in ABC can likewise be interpreted through suitable loss functions. There is, however, an important distinction. In ABC, these weights typically arise from acceptance probabilities involving simulated data and a prescribed tolerance. Their probabilistic origin introduces questions concerning the behavior of the posterior as the tolerance or sample size changes. Variational principles provide a complementary way to study these questions, rather than an alternative definition of the ABC posterior.

The purpose of the present paper is to investigate these connections within a common measure-theoretic formulation. We describe compatibility between observed data and candidate statistical models through nonnegative weights defined on the parameter space. After normalization with respect to the prior, these weights determine a family of posterior distributions. This formulation accommodates standard ABC acceptance probabilities while allowing the convergence mechanism to be studied independently of a particular simulation procedure. Two perspectives then emerge. The first concerns the variational characterization of compatibility-based posteriors through entropy-regularized minimization. The second focuses on the geometry induced by Wasserstein discrepancies in the large-sample regime.

We begin by examining the regime in which the compatibility threshold approaches zero. The main question is whether the corresponding posterior distributions admit a well-defined limit and how this limit depends on the compatibility weights. We show that convergence in mean of suitably normalized compatibility weights with respect to the prior ensures convergence of the posterior distributions in total variation. The limiting posterior is determined by the resulting compatibility profile. Although the underlying normalization argument is classical, this formulation makes explicit which properties of the weights are sufficient for posterior convergence. It also separates these properties from the specific discrepancy and acceptance mechanism used to construct the posterior.

We next investigate the large-sample regime. In this regime, the asymptotic discrepancy is determined

by the limiting observational distribution and the sampling distribution associated with each parameter value. This discrepancy identifies the models that can approximate the observational law at a prescribed tolerance. Its essential infimum under the prior determines a critical compatibility level, reflecting the smallest discrepancy attainable among parameter values that are relevant to the prior.

For a fixed tolerance above this critical level, we establish a compatibility-filtering result. Under suitable assumptions, the posterior converges in probability, with respect to total variation, to the prior conditioned on the region where the limiting discrepancy is strictly below the tolerance. The interpretation is as follows: as the sample size increases, parameter values outside this region are progressively excluded, while the relative prior weights within the region are preserved. Consequently, the limiting posterior need not concentrate on the minimizers of the discrepancy. Even when this discrepancy has a unique minimizer, a positive tolerance may leave a nontrivial region of compatible parameter values.

The analysis also considers situations in which a complete filtering limit cannot be established. Under weaker forms of asymptotic stabilization, we derive lower bounds on acceptance probabilities and prior-averaged compatibility. These results distinguish the identification of a limiting posterior from the persistence of compatibility within prescribed parameter regions. The critical threshold depends on the prior and need not coincide with the minimum discrepancy over the full parameter space.

The second part of the paper develops a variational interpretation of posterior distributions arising from compatibility-based inference. A compatibility weight defines a loss through its negative logarithm. Combining this loss with the Kullback–Leibler divergence from the prior leads to an entropy-regularized minimization problem. By the classical Gibbs–Donsker–Varadhan variational principle, the corresponding posterior is the unique minimizer of this problem; see, e.g., [10, 11]. This provides a precise variational characterization of the posterior without altering its probabilistic construction.

We use this representation to investigate variational stability as the compatibility threshold vanishes. The limiting compatibility profile determines an optimization problem whose unique minimizer is approached by every family of asymptotic minimizers of the threshold-dependent problems. Thus, convergence of compatibility weights ensures both convergence of posterior distributions and stability of the associated variational problems. The contribution lies in establishing this connection between posterior asymptotics and variational stability, rather than in the classical Gibbs principle.

We then specialize the asymptotic discrepancy to a Wasserstein distance. In this case, each parameter value determines a probability distribution, and the comparison with the limiting observational law takes place in a Wasserstein space. This suggests shifting attention from the parameters to the distributions generated by the statistical model. A probability measure on the parameter space induces, through the parameter-to-distribution map, a probability measure on the space of model laws. The compatibility functional can therefore be expressed directly in terms of this induced measure.

A Wasserstein ball centered at the limiting observational distribution describes the model laws whose discrepancy does not exceed the critical threshold. The compatibility cost measures departures from this set. For suitable cost functions, the associated functional admits a representation in terms of expected powers of the distance from the ball. This makes the role of the threshold explicit and provides a geometric interpretation of the penalty assigned to incompatible model laws.

The transport formulation also leads to an exact variational reduction. When the compatibility cost depends on a parameter only through its sampling distribution, the entropy-regularized minimization problem can be reformulated in terms of probability distributions over model laws. We show that this reformulation preserves the minimum value and establishes a correspondence between minimizers through the same parameter-to-distribution map. The result follows from the data-processing inequality for relative entropy, together with an explicit construction that attains the corresponding lower bound.

Together, these results connect three aspects of discrepancy-based inference. The asymptotic analysis identifies parameter values compatible with the observational law. The variational formulation characterizes posterior distributions through entropy-regularized optimization. The transport representation expresses compatibility costs through the geometry of probability measures. The aim is not to propose an alternative updating rule, but to identify the mathematical structures underlying compatibility-based inference.

Selected aspects of the theory are illustrated through applications. Gaussian distributions provide explicit Wasserstein formulas and a numerical example of compatibility filtering under model misspecification. Finite-state models lead to finite-dimensional transport problems, while mixed discrete–continuous and

deterministic ones demonstrate the flexibility of the distributional formulation. These examples help distinguish the geometry of compatibility from the representation of the parameter space.

The remainder of the paper is organized as follows. Section 1 establishes the measure-theoretic foundations of ABC and derives the corresponding posterior construction. Section 2 introduces a general compatibility-weight formulation and studies vanishing-threshold limits. Section 3 develops the asymptotic compatibility theory, including fixed-threshold filtering and persistence results. Section 4 examines the resulting variational and transport structures, together with their interpretation in terms of incompatibility costs. Finally, Section 5 summarizes the main findings and outlines directions for future research.

1 Approximate Bayesian Computation

This section introduces the observation and parameter spaces, the underlying probabilistic model, and the discrepancy-based construction of the corresponding posterior distributions.

1.1 Probabilistic foundations of ABC

Fix a probability space $(\Omega, \mathcal{A}, \mathbf{P})$. For any topological space X , we denote by $\mathcal{B}(X)$ its Borel σ -algebra and by $\mathcal{P}(X)$ the set of probability measures defined on $\mathcal{B}(X)$. Given another topological space Y , $\mathcal{B}(X, Y)$ stands for the class of Borel measurable mappings from X to Y .

Let

$$\mathcal{Y} \subseteq \mathbb{R}^{m_Y} \equiv \mathbb{R}^{m_Y \times 1}$$

be the observation space endowed with the topological structure induced by a metric d_Y , where

$$m_Y \in \mathbb{N}^* := \mathbb{N} \setminus \{0\}$$

gives the dimension of a generic data point $\mathbf{y} \in \mathcal{Y}$. We also specify the sample size $n \in \mathbb{N}^*$.

The observed random sample is represented by the random vector

$$\mathbf{Y}^{1:n} \equiv [\mathbf{Y}^1, \dots, \mathbf{Y}^n]: (\Omega, \mathcal{A}) \longrightarrow (\mathcal{Y}^n, \mathcal{B}(\mathcal{Y}^n)),$$

with distribution $\mathbf{P}^{1:n} \in \mathcal{P}(\mathcal{Y}^n)$.

Thus, to every elementary outcome $\omega \in \Omega$ there corresponds a finite sequence of observed quantities: for every index $i = 1, \dots, n$, the i -th observation corresponding to ω is denoted by

$$\mathbf{y}^i := \mathbf{Y}^i(\omega) \in \mathcal{Y}.$$

These observations are collectively arranged into the sample vector at ω , that is,

$$\mathbf{y}^{1:n} := \mathbf{Y}^{1:n}(\omega) \in \mathcal{Y}^n \subseteq \mathbb{R}^{m_Y \times n}.$$

In what follows, the dependence upon ω will be tacitly omitted whenever no ambiguity arises, so that the notation $\mathbf{y}^{1:n} \in \mathcal{Y}^n$ is understood as a fixed realization of the observed random sample.

We next introduce the parameter space

$$\Theta \subseteq \mathbb{R}^{m_\Theta},$$

governing the data-generating mechanism and equipped with the topology induced by a metric d_Θ , where

$$m_\Theta \in \mathbb{N}^*$$

is the dimension of the parameter vectors $\boldsymbol{\vartheta} \in \Theta$.

We further consider a prior distribution

$$\pi \in \mathcal{P}(\Theta)$$

on the parameter space Θ , together with a Markov kernel from Θ to $\mathcal{Y}^n$,

$$K_n: \Theta \times \mathcal{B}(\mathcal{Y}^n) \longrightarrow [0, 1],$$

defining the associated family of sampling distributions: for every $\vartheta \in \Theta$, the probability measure

$$\mu_{\vartheta}^n := K_n(\vartheta, \cdot) \in \mathcal{P}(\mathcal{Y}^n)$$

represents the distribution of a pseudo-sample generated under the parameter ϑ .

In particular, for every $D \in \mathcal{B}(\mathcal{Y}^n)$, the mapping

$$\vartheta \longmapsto \mu_{\vartheta}^n[D]$$

belongs to $\mathcal{B}(\Theta, [0, 1])$ and is therefore π -integrable (in the Lebesgue sense).

The first structural assumption underlying Approximate Bayesian Computation can be stated and will henceforth be maintained throughout the paper.

(A-1) — *Simulatability*. The statistical model $\{\mu_{\vartheta}^n\}_{\vartheta \in \Theta}$ induced by K_n is generative: for every $\vartheta \in \Theta$, one can simulate arbitrarily many independent pseudo-samples

$$\mathbf{Y}_{\vartheta}^{1:n} \equiv [Y_{\vartheta}^1, \dots, Y_{\vartheta}^n] \in \mathcal{Y}^n,$$

namely random vectors distributed according to μ_{ϑ}^n ; that is,

$$\mathbf{Y}_{\vartheta}^{1:n} \sim \mu_{\vartheta}^n.$$

For every $\vartheta \in \Theta$, any realization of the random pseudo-sample $\mathbf{Y}_{\vartheta}^{1:n}$ is written as

$$\mathbf{y}_{\vartheta}^{1:n} \in \mathcal{Y}^n.$$

We emphasize that the observed sample vector $\mathbf{y}^{1:n}$ constitutes the fixed empirical realization arising from the underlying statistical experiment and is regarded as given, whereas any pseudo-sample $\mathbf{y}_{\vartheta}^{1:n}$ is a realization of simulated data generated under a prescribed parameter value $\vartheta \in \Theta$.

This distinction lies at the heart of ABC: in the absence of direct likelihood evaluation, inference compares observed and simulated data through a discrepancy measure

$$\mathcal{D}: \mathcal{Y}^n \times \mathcal{Y}^n \longrightarrow \mathbb{R}_+ := [0, \infty[,$$

defined as a continuous pseudo-metric on $\mathcal{Y}^n$, satisfying all the standard axioms of a metric except that distinct elements may have zero distance; see, e.g., [34, 4, 30].

To this end, for any observed sample $\mathbf{y}^{1:n} \in \mathcal{Y}^n$ and radius $r \in \mathbb{R}_+$, we introduce the sample-dependent closed subset $D_r(\mathbf{y}^{1:n})$ of the observation space $\mathcal{Y}^n$ containing all elements whose deviation from $\mathbf{y}^{1:n}$ does not exceed r under the discrepancy measure $\mathcal{D}$; namely,

$$D_r(\mathbf{y}^{1:n}) := \mathcal{D}(\mathbf{y}^{1:n}, \cdot)^{-1}([0, r]) = \{\tilde{\mathbf{y}}^{1:n} \in \mathcal{Y}^n \mid \mathcal{D}(\mathbf{y}^{1:n}, \tilde{\mathbf{y}}^{1:n}) \leq r\} \in \mathcal{B}(\mathcal{Y}^n).$$

The family $\{D_r(\mathbf{y}^{1:n})\}_{r \geq 0}$ is increasing in r and, as $r \downarrow 0$,

$$D_r(\mathbf{y}^{1:n}) \downarrow D_0(\mathbf{y}^{1:n}) \equiv \mathcal{D}(\mathbf{y}^{1:n}, \cdot)^{-1}(\{0\}).$$

We are in a position to formulate the second fundamental assumption of Approximate Bayesian Computation, which will be adopted from this point onward.

(A-2) — *Nondegeneracy*. There exists $\varepsilon^\circ \equiv \varepsilon_n^\circ \in]0, \infty[$ such that, for every $\varepsilon \in]0, \varepsilon^\circ[$ —hereafter referred to as an ABC threshold—and for $\mathbf{P}^{1:n}$ -almost every $\mathbf{y}^{1:n} \in \mathcal{Y}^n$,

$$C_\varepsilon^n(\mathbf{y}^{1:n}) := \int_{\Theta} \mu_{\vartheta}^n[D_\varepsilon(\mathbf{y}^{1:n})] \pi(d\vartheta) > 0. \quad (1)$$

By monotonicity in ε , the exceptional null set in (A-2) can be chosen independently of the threshold, ensuring that the stated positivity holds simultaneously for all $\varepsilon \in]0, \varepsilon^\circ[$.

For each fixed $\mathbf{y}^{1:n} \in \mathcal{Y}^n$, ABC threshold ε , and $\vartheta \in \Theta$, the quantity $\mu_\vartheta^n[D_\varepsilon(\mathbf{y}^{1:n})]$ represents the acceptance probability under the model with parameter ϑ , namely the probability that simulated data generated under ϑ fall within the ε -neighborhood $D_\varepsilon(\mathbf{y}^{1:n})$ of the empirical realization $\mathbf{y}^{1:n}$.

As a function of the parameter ϑ , the acceptance probability

$$\vartheta \mapsto \mu_\vartheta^n[D_\varepsilon(\mathbf{y}^{1:n})] \quad (2)$$

quantifies the compatibility between the model under ϑ and the observed sample, and may be interpreted as a surrogate likelihood, in which the classical pointwise comparison between observed and model-implied data is replaced by a tolerance-based notion of proximity; see, e.g., [24, 37]. Large values indicate that the model under ϑ assigns high probability to simulated realizations close to the observed sample, while small values reflect poor compatibility with the observed data.

1.2 ABC posterior distribution

We now turn to the probabilistic formulation of the ABC posterior distribution. When direct likelihood evaluation is unavailable or computationally intractable, but simulation from the statistical model remains feasible according to (A-1), approximate Bayesian inference can be performed by reweighting the prior distribution π according to the compatibility between observed and simulated data.

Given an ABC threshold ε , each parameter value $\vartheta \in \Theta$ is weighted according to the corresponding acceptance probability. The resulting normalized measure defines an approximate Bayesian update and is formalized below as the ABC posterior distribution, in which the role traditionally played by the likelihood is taken precisely by the compatibility map (2); see, e.g., [24, 37].

The following scheme summarizes the classical ABC rejection algorithm; see, e.g., [3, 24].

Algorithm 1 (ABC rejection algorithm). *Let $\mathbf{y}^{1:n} \in \mathcal{Y}^n$ be an observed sample.*

Step 1 (Tolerance specification). *Choose an ABC threshold $\varepsilon \in]0, \varepsilon^\circ[$.*

Step 2 (Sampling and simulation). *Draw a parameter $\vartheta \in \Theta$ from the prior distribution π and generate a pseudo-sample $\mathbf{y}_\vartheta^{1:n}$ according to μ_ϑ^n .*

Step 3 (Acceptance rule). *Retain ϑ if and only if $\mathbf{y}_\vartheta^{1:n} \in D_\varepsilon(\mathbf{y}^{1:n})$.*

For every ABC threshold ε , consider the ABC normalizing function defined on $\mathcal{Y}^n$ through (1),

$$\mathbf{y}^{1:n} \mapsto C_\varepsilon^n(\mathbf{y}^{1:n}), \quad \mathbf{y}^{1:n} \in \mathcal{Y}^n.$$

By (A-2), $C_\varepsilon^n(\cdot) > 0$ $\mathbf{P}^{1:n}$ -a.s. Moreover, $C_\varepsilon^n(\cdot) \in \mathcal{B}(\mathcal{Y}^n, [0, 1])$ since the Markov-kernel property of K_n , together with the assumed continuity of $\mathcal{D}$, ensures the Borel measurability of the mapping

$$(\vartheta, \mathbf{y}^{1:n}) \mapsto \mu_\vartheta^n[D_\varepsilon(\mathbf{y}^{1:n})], \quad (\vartheta, \mathbf{y}^{1:n}) \in \Theta \times \mathcal{Y}^n.$$

It follows that the function $\Pi_\varepsilon^n: \mathcal{Y}^n \times \mathcal{B}(\Theta) \rightarrow [0, 1]$ given, for every $\mathbf{y}^{1:n} \in \mathcal{Y}^n$ and $B \in \mathcal{B}(\Theta)$, by

$$\Pi_\varepsilon^n(\mathbf{y}^{1:n}, B) := \begin{cases} \frac{1}{C_\varepsilon^n(\mathbf{y}^{1:n})} \int_B \mu_\vartheta^n[D_\varepsilon(\mathbf{y}^{1:n})] \pi(d\vartheta), & \text{if } C_\varepsilon^n(\mathbf{y}^{1:n}) > 0, \\ \pi[B], & \text{otherwise,} \end{cases}$$

is a Markov kernel from $\mathcal{Y}^n$ to Θ , and

$$\Pi_\varepsilon^n(\mathbf{Y}^{1:n}, \cdot)$$

is the random ABC posterior distribution induced by the observed sample.

For every fixed $\mathbf{y}^{1:n} \in \mathcal{Y}^n$ with $C_\varepsilon^n(\mathbf{y}^{1:n}) > 0$, we write the corresponding posterior distribution as

$$\pi_\varepsilon^n \equiv \pi_\varepsilon^{\mathbf{y}^{1:n}} := \Pi_\varepsilon^n(\mathbf{y}^{1:n}, \cdot) \in \mathcal{P}(\Theta). \quad (3)$$

By construction,

$$\pi_\varepsilon^n \ll \pi$$

and its Radon–Nikodym derivative with respect to π is proportional to the mapping (2).

The foregoing construction shows that repeated independent executions of steps 2–3 of Algorithm 1 generate accepted parameter values distributed according to π_ε^n .

2 On the vanishing-threshold limit

In this section, we study the asymptotic behavior of the posterior distributions (3) as the tolerance parameter ε vanishes, with the sample size n held fixed. We introduce a general formulation in terms of nonnegative compatibility weights and establish sufficient conditions for convergence in total variation.

The complementary large-sample regime is investigated separately in Section 3.

2.1 A general convergence principle

Consider a general family of nonnegative measurable functions

$$w_\varepsilon \in \mathcal{B}(\Theta, \mathbb{R}_+),$$

indexed by $\varepsilon \in]0, \infty[$, intended to quantify the compatibility between the model under a parameter value $\vartheta \in \Theta$ and the observed sample $\mathbf{y}^{1:n}$, in the same spirit as (2); we refer to these functions as compatibility weights. We assume that, for every sufficiently small ε , w_ε is π -integrable and not equal to zero π -almost everywhere. Accordingly, these functions induce the family $\{\pi_\varepsilon\}_\varepsilon$ in $\mathcal{P}(\Theta)$ defined by

$$\pi_\varepsilon[B] := \frac{\int_B w_\varepsilon(\vartheta) \pi(d\vartheta)}{\int_\Theta w_\varepsilon(\vartheta') \pi(d\vartheta')}, \quad B \in \mathcal{B}(\Theta). \quad (4)$$

This formulation encompasses the classical ABC posterior (3) as a special case; see, e.g., [3, 24, 37].

We next establish a fundamental convergence principle governed by the asymptotic behavior of the compatibility weights w_ε in the vanishing-threshold regime.

Proposition 2.1. *Assume that there exist a family $\{\kappa_\varepsilon\}_\varepsilon$ of strictly positive constants and a function $\bar{w} \in \mathcal{B}(\Theta, \mathbb{R}_+)$, π -integrable and not π -almost everywhere equal to zero, such that*

$$\lim_{\varepsilon \downarrow 0} \int_\Theta \left| \frac{1}{\kappa_\varepsilon} w_\varepsilon(\vartheta) - \bar{w}(\vartheta) \right| \pi(d\vartheta) = 0. \quad (5)$$

Then, as $\varepsilon \downarrow 0$, the family of posterior distributions $\{\pi_\varepsilon\}_\varepsilon$ introduced in (4) converges in total variation to the probability measure $\bar{\pi} \in \mathcal{P}(\Theta)$ defined by

$$\bar{\pi}[B] := \frac{\int_B \bar{w}(\vartheta) \pi(d\vartheta)}{\int_\Theta \bar{w}(\vartheta') \pi(d\vartheta')}, \quad B \in \mathcal{B}(\Theta), \quad (6)$$

that is,

$$\lim_{\varepsilon \downarrow 0} \sup_{B \in \mathcal{B}(\Theta)} |\pi_\varepsilon[B] - \bar{\pi}[B]| = 0. \quad (7)$$

Proof. For every $B \in \mathcal{B}(\Theta)$, by the definitions above and the triangle inequality for integrals,

$$\begin{aligned} & |\pi_\varepsilon[B] - \bar{\pi}[B]| \\ &= \left| \frac{\int_B \left[\frac{1}{\kappa_\varepsilon} w_\varepsilon(\vartheta) - \bar{w}(\vartheta) \right] \pi(d\vartheta)}{\int_\Theta \frac{1}{\kappa_\varepsilon} w_\varepsilon(\vartheta') \pi(d\vartheta')} + \left[\frac{1}{\int_\Theta \frac{1}{\kappa_\varepsilon} w_\varepsilon(\vartheta') \pi(d\vartheta')} - \frac{1}{\int_\Theta \bar{w}(\vartheta') \pi(d\vartheta')} \right] \int_B \bar{w}(\vartheta) \pi(d\vartheta) \right| \\ &\leq \frac{\int_\Theta \left| \frac{1}{\kappa_\varepsilon} w_\varepsilon(\vartheta) - \bar{w}(\vartheta) \right| \pi(d\vartheta)}{\int_\Theta \frac{1}{\kappa_\varepsilon} w_\varepsilon(\vartheta') \pi(d\vartheta')} + \left| \frac{1}{\int_\Theta \frac{1}{\kappa_\varepsilon} w_\varepsilon(\vartheta') \pi(d\vartheta')} - \frac{1}{\int_\Theta \bar{w}(\vartheta') \pi(d\vartheta')} \right| \int_\Theta \bar{w}(\vartheta) \pi(d\vartheta). \end{aligned}$$

Both terms on the right-hand side converge to zero as $\varepsilon \downarrow 0$ directly from (5). Since the bound is independent of $B \in \mathcal{B}(\Theta)$, taking the supremum proves (7). $\square$

Within the present framework, Proposition 2.1 highlights that the asymptotic behavior of the induced posterior measures π_ε is essentially governed by the corresponding rescaled compatibility weights $w_\varepsilon/\kappa_\varepsilon$. In particular, convergence in integral form of such weights— $L^1(\pi)$ -convergence—as specified in (5) yields convergence in total variation toward a well-defined limiting posterior distribution $\bar{\pi}$.

This compatibility-weight formulation also underlies the variational and transport formulations of approximate Bayesian inference developed in Section 4.

2.2 Bayesian posterior recovery in classical ABC

The preceding approach applies to a broad class of approximate Bayesian constructions. We first specialize it to the ABC setting based on the full sample, where the compatibility weights are induced by acceptance probabilities over shrinking neighborhoods of the observed realization; see, e.g., [3, 24, 37].

For this purpose, let $\mathbf{y}^{1:n} \in \mathcal{Y}^n$ be the observed sample and assume that there exists a σ -finite measure λ^n on $\mathcal{B}(\mathcal{Y}^n)$ such that, for π -almost every $\vartheta \in \Theta$, the probability measure μ_ϑ^n is absolutely continuous with respect to λ^n , with Radon–Nikodym derivative

$$f_\vartheta^n := \frac{d\mu_\vartheta^n}{d\lambda^n}$$

admitting a version, still denoted by f_ϑ^n , which is continuous at $\mathbf{y}^{1:n}$. The generality of the dominating measure λ^n allows for continuous, discrete, lower-dimensional, and mixed data, including settings described by Lebesgue, counting, Hausdorff, or suitable combinations of such measures.

Moreover, define

$$\bar{w}(\vartheta) := f_\vartheta^n(\mathbf{y}^{1:n}) \quad (8)$$

and suppose that it belongs to $\mathcal{B}(\Theta, \mathbb{R}_+)$ and is not π -almost everywhere equal to zero. The function (8) then provides a version of the likelihood associated with $\mathbf{y}^{1:n}$. Accordingly, for every ABC threshold ε , set

$$\kappa_\varepsilon := \lambda^n[D_\varepsilon(\mathbf{y}^{1:n})]$$

and, for π -almost every $\vartheta \in \Theta$,

$$w_\varepsilon(\vartheta) := \mu_\vartheta^n[D_\varepsilon(\mathbf{y}^{1:n})] = \int_{D_\varepsilon(\mathbf{y}^{1:n})} f_\vartheta^n(\mathbf{z}^{1:n}) \lambda^n(d\mathbf{z}^{1:n}).$$

Assume, in addition, that $\mathcal{D}$ is a genuine metric compatible with the topology of $\mathcal{Y}^n$, and that

$$0 < \kappa_\varepsilon < \infty$$

for every sufficiently small ε . Then, for π -almost every $\vartheta \in \Theta$, continuity of f_ϑ^n at $\mathbf{y}^{1:n}$ yields, as $\varepsilon \downarrow 0$,

$$\left| \frac{1}{\kappa_\varepsilon} w_\varepsilon(\vartheta) - \bar{w}(\vartheta) \right| \leq \sup_{\mathbf{z}^{1:n} \in D_\varepsilon(\mathbf{y}^{1:n})} |f_\vartheta^n(\mathbf{z}^{1:n}) - f_\vartheta^n(\mathbf{y}^{1:n})| \longrightarrow 0.$$

Therefore, if there exists a function $g \in \mathcal{B}(\Theta, \mathbb{R}_+)$ which is π -integrable and satisfies, for every sufficiently small ε and for π -almost every $\vartheta \in \Theta$,

$$\frac{1}{\kappa_\varepsilon} w_\varepsilon(\vartheta) \leq g(\vartheta),$$

then the integral convergence (5) follows from the dominated convergence theorem. Consequently, Proposition 2.1 implies convergence in total variation, as $\varepsilon \downarrow 0$, of the ABC posterior $\pi_\varepsilon \equiv \pi_\varepsilon^n$ toward the probability measure (6), here explicitly given by

$$\bar{\pi}[B] = \frac{\int_B f_\vartheta^n(\mathbf{y}^{1:n}) \pi(d\vartheta)}{\int_\Theta f_{\vartheta'}^n(\mathbf{y}^{1:n}) \pi(d\vartheta')}, \quad B \in \mathcal{B}(\Theta).$$

From a measure-theoretic perspective, $\bar{\pi}$ provides a version of the usual likelihood-based posterior distribution given the full observed sample $\mathbf{y}^{1:n}$, thus recovering the classical Bayesian update. Its density with respect to π is proportional to the likelihood function (8); see, e.g., [33].

Remark 2.1. The preceding recovery of the usual Bayesian posterior relies on comparing the full observed sample with simulated data. More generally, let $(\mathcal{S}, d_S)$ be a metric space and suppose that the discrepancy is induced by a measurable summary map $S^n: \mathcal{Y}^n \rightarrow \mathcal{S}$ through

$$\mathcal{D}(\mathbf{y}^{1:n}, \mathbf{z}^{1:n}) = d_S(S^n(\mathbf{y}^{1:n}), S^n(\mathbf{z}^{1:n})).$$

Under analogous regularity and domination conditions for the summary distributions, the limiting posterior is determined by the likelihood associated with $S^n(\mathbf{y}^{1:n})$; see, e.g., [14, 24]. Hence, it coincides with the posterior distribution given the observed summary statistic, which generally differs from the posterior given the full sample; the two coincide, in particular, when S^n is sufficient.

When $\mathcal{D}$ is only a pseudo-metric, observations in the equivalence class $\mathcal{D}(\mathbf{y}^{1:n}, \cdot)^{-1}(\{0\})$ cannot be distinguished as $\varepsilon \downarrow 0$, so that the limiting information need not identify the single realization $\mathbf{y}^{1:n}$.

The previous analysis shows that classical ABC convergence arises as a particular instance of Proposition 2.1. The fundamental mechanism underlying posterior convergence is not tied to the structure of ABC neighborhoods, but rather to integral control of suitably normalized compatibility weights.

Within this framework, uniform local bounds on the likelihood are not intrinsically required, as the necessary integral convergence may instead be obtained through an integrated domination condition with respect to the prior distribution. Thus, regularity assumptions appearing in classical ABC convergence analyses (see, e.g., [24, 37]) may be viewed as sufficient means of establishing such control, rather than as intrinsic requirements of the general convergence principle.

3 The large-sample regime

We now examine the large-sample behavior of posterior distributions through a compatibility structure induced on the parameter space. We introduce a limiting discrepancy profile and compatibility regions, which identify the parameter values compatible with the limiting observational law.

For a fixed tolerance above the critical threshold, we establish conditions under which the posterior converges to the prior conditioned on the corresponding compatibility region. We also derive persistence results under weaker asymptotic assumptions. These findings distinguish compatibility filtering from posterior concentration in the classical parametric sense. In particular, concentration on the minimizers of the limiting discrepancy is not guaranteed, even as the sample size tends to infinity. This distinction is relevant under model misspecification or insufficient identifiability; see, e.g., [16, 17].

3.1 Probabilistic representations and asymptotic discrepancies

The construction below separates the sampling mechanism of the ABC procedure from the asymptotic discrepancy geometry governing acceptance. Rather than working with finite-dimensional sample vectors, we represent observed and simulated samples by probability measures. This representation is well suited to the large-sample regime, since the ambient spaces vary with n , while these probability measures take values in a common space. Acceptance is then expressed in terms of the discrepancy between them.

We equip $\mathcal{P}(\mathcal{Y})$ with a generic pseudo-metric

$$\mathcal{T}: \mathcal{P}(\mathcal{Y}) \times \mathcal{P}(\mathcal{Y}) \longrightarrow \mathbb{R}_+. \quad (9)$$

This level of generality naturally encompasses transport-based discrepancies, whose role in statistical inference will be further explored in Section 4; see, e.g., [4].

For every sample size $n \in \mathbb{N}^*$, let

$$\nu_n: \mathcal{Y}^n \longrightarrow \mathcal{P}(\mathcal{Y})$$

and, for every $\vartheta \in \Theta$,

$$\nu_{\vartheta,n}: \mathcal{Y}^n \longrightarrow \mathcal{P}(\mathcal{Y})$$

denote probabilistic representation maps for the observed and simulated samples, respectively. More explicitly, any realization $\mathbf{y}^{1:n} \in \mathcal{Y}^n$ of the observed random sample and any pseudo-sample $\mathbf{y}_\vartheta^{1:n} \in \mathcal{Y}^n$ generated under parameter ϑ are respectively associated with the probability measures

$$\nu_n(\mathbf{y}^{1:n}) \in \mathcal{P}(\mathcal{Y})$$

and

$$\nu_{\vartheta,n}(\mathbf{y}_\vartheta^{1:n}) \in \mathcal{P}(\mathcal{Y}).$$

A natural choice is given by empirical measures, although the subsequent analysis does not rely on this specific representation; see, e.g., [4, 32]. What is essential is that observed and simulated samples are mapped into the common space $\mathcal{P}(\mathcal{Y})$, so that the asymptotic behavior of the ABC acceptance mechanism can be characterized in terms of the geometry induced by $\mathcal{T}$.

Remark 3.1. Since $\mathcal{T}$ is only assumed to be a pseudo-metric, the topology induced on $\mathcal{P}(\mathcal{Y})$ need not be Hausdorff (equivalently, T_2). In particular, distinct probability measures on $\mathcal{Y}$ may be topologically indistinguishable whenever their discrepancy with respect to $\mathcal{T}$ vanishes.

We next assume that, for every $n \in \mathbb{N}^*$, the mapping

$$(\vartheta, \mathbf{y}^{1:n}, \mathbf{y}_\vartheta^{1:n}) \mapsto \Delta_{\vartheta,n}(\mathbf{y}^{1:n}, \mathbf{y}_\vartheta^{1:n}) := \mathcal{T}(\nu_n(\mathbf{y}^{1:n}), \nu_{\vartheta,n}(\mathbf{y}_\vartheta^{1:n})) \quad (10)$$

belongs to $\mathcal{B}(\Theta \times \mathcal{Y}^n \times \mathcal{Y}^n, \mathbb{R}_+)$.

The large-sample behavior of the ABC compatibility selection mechanism is ultimately governed by the discrepancy geometry emerging in $\mathcal{P}(\mathcal{Y})$, rather than by finite-dimensional sample representations. To identify this asymptotic structure, we assume the existence of a probability measure

$$\mu^\star \in \mathcal{P}(\mathcal{Y}) \quad (11)$$

such that, for every $n \in \mathbb{N}^*$, the mapping

$$\omega \mapsto \mathcal{T}_n^\star(\omega) := \mathcal{T}(\mu^\star, \nu_n(\mathbf{Y}^{1:n}(\omega))) \quad (12)$$

is $\mathcal{A}$ -measurable and converges to zero in $\mathbf{P}$ -probability as $n \rightarrow \infty$, namely

$$\mathcal{T}_n^\star \xrightarrow[n \rightarrow \infty]{\mathbf{P}} 0. \quad (13)$$

This condition is fulfilled, in particular, whenever $\mathcal{T}_n^\star$ converges to zero $\mathbf{P}$ -almost surely.

Similarly, for every $\vartheta \in \Theta$, we assume the existence of a probability measure

$$\nu_\vartheta \in \mathcal{P}(\mathcal{Y}) \quad (14)$$

such that the function

$$\vartheta \mapsto \mathcal{T}_\vartheta^\star := \mathcal{T}(\mu^\star, \nu_\vartheta) \quad (15)$$

belongs to $\mathcal{B}(\Theta, \mathbb{R}_+)$ and, for π -almost every $\vartheta \in \Theta$ and every $n \in \mathbb{N}^*$, the mapping

$$\mathbf{y}_\vartheta^{1:n} \mapsto \mathcal{T}(\nu_{\vartheta,n}(\mathbf{y}_\vartheta^{1:n}), \nu_\vartheta)$$

belongs to $\mathcal{B}(\mathcal{Y}^n, \mathbb{R}_+)$ and, for every $\xi > 0$,

$$\lim_{n \rightarrow \infty} \mu_\vartheta^n \left[\left\{ \mathbf{y}_\vartheta^{1:n} \in \mathcal{Y}^n \mid \mathcal{T}(\nu_{\vartheta,n}(\mathbf{y}_\vartheta^{1:n}), \nu_\vartheta) > \xi \right\} \right] = 0. \quad (16)$$

This amounts to convergence in probability under the sequence of pseudo-sample laws $\{\mu_\vartheta^n\}_{n \in \mathbb{N}^*}$.

The quantity $\mathcal{T}_\vartheta^\star$ represents the asymptotic discrepancy level of ϑ , comparing its model law with the limiting observational law. As ϑ varies over Θ , the function (15) induces the asymptotic compatibility geometry of the model class and determines the relative discrepancy levels across parameter values.

Although the limiting probability measures need not be uniquely determined under a pseudo-metric, the asymptotic discrepancy levels defined in (15) are uniquely determined up to π -null sets.

The asymptotic discrepancy function (15) determines two thresholds, according to whether it is assessed relative to the prior or over the entire model class: the π -essential compatibility threshold

$$\varepsilon_\pi^\star := \operatorname{ess\,inf}_{\vartheta \in \Theta} \mathcal{T}_\vartheta^\star, \quad (17)$$

where the essential infimum is taken with respect to π , and the model-wide compatibility threshold

$$\varepsilon_\Theta^\star := \inf_{\vartheta \in \Theta} \mathcal{T}_\vartheta^\star \leq \varepsilon_\pi^\star. \quad (18)$$

The threshold $\varepsilon_\pi^\star$ represents the essential lower bound of the asymptotic discrepancy under π , while $\varepsilon_\Theta^\star$ describes the best level available over the model class. Strict inequality in (18) thus reflects a restriction induced by the prior relative to the discrepancy geometry. In particular, $\varepsilon_\Theta^\star = 0$ means that the limiting observational law can be matched arbitrarily closely within the model class; if $\varepsilon_\Theta^\star > 0$, an intrinsic asymptotic discrepancy remains, as may occur under model misspecification, structural incompatibility, or limited expressive capacity of the simulated probabilistic family; see, e.g., [17].

For a fixed effective ABC tolerance $\tau \in]0, \infty[$, the asymptotic behavior of the compatibility selection mechanism depends on its relation to the threshold $\varepsilon_\pi^\star$. We restrict attention to the supercritical regime

$$\tau > \varepsilon_\pi^\star. \quad (19)$$

The boundary case $\tau = \varepsilon_\pi^\star$ may depend on prior mass near this threshold and on sampling fluctuations, requiring a local asymptotic analysis. In the subcritical regime $\tau < \varepsilon_\pi^\star$, acceptance probabilities vanish asymptotically for π -almost every parameter. The normalized posterior may then be governed by their relative decay rates, calling for large-deviation or Laplace-type analysis.

Remark 3.2. The assumptions introduced above can be satisfied in a broad range of standard Approximate Bayesian Computation settings. This covers discrepancy constructions induced by Wasserstein distances and kernel-based metrics such as the maximum mean discrepancy; see, e.g., [38, 34, 4]. Functionals built from measurable summary statistics may also be accommodated, as may other probability metrics, including Prokhorov or total variation discrepancies, whenever the chosen probabilistic representations satisfy the required measurability and convergence properties.

The probability measures $\mu^\star$ and ν_ϑ may arise as limits under familiar probabilistic regimes, including i.i.d. sampling, ergodic observations, and weak-convergence settings, provided that $\mathcal{T}$ is compatible with the mode of convergence. For empirical measures in Wasserstein distance, see, e.g., [6, 15].

3.2 Acceptance geometry and posterior compatibility

The discrepancy map (10) determines the finite-sample acceptance mechanism. Consistently with Algorithm 1, a pseudo-sample $\mathbf{y}_\vartheta^{1:n}$ generated under $\vartheta \in \Theta$ is retained whenever the discrepancy between $\nu_{\vartheta,n}(\mathbf{y}_\vartheta^{1:n})$ and the observed representation $\nu_n(\mathbf{y}^{1:n})$ does not exceed the tolerance τ .

For every $n \in \mathbb{N}^*$ and $\vartheta \in \Theta$, define the data-dependent finite-sample acceptance region

$$A_{\vartheta,n}(\mathbf{Y}^{1:n}; \tau) := \{ \mathbf{y}_\vartheta^{1:n} \in \mathcal{Y}^n \mid \Delta_{\vartheta,n}(\mathbf{Y}^{1:n}, \mathbf{y}_\vartheta^{1:n}) \leq \tau \}. \quad (20)$$

By measurability of $\Delta_{\vartheta,n}$, for every $\omega \in \Omega$,

$$A_{\vartheta,n}(\mathbf{Y}^{1:n}(\omega); \tau) \in \mathcal{B}(\mathcal{Y}^n).$$

Consistently with Section 2, the finite-sample compatibility weight associated with $\vartheta \in \Theta$ is defined as

$$w_\tau^n(\vartheta; \mathbf{Y}^{1:n}) := \mu_\vartheta^n[A_{\vartheta,n}(\mathbf{Y}^{1:n}; \tau)]. \quad (21)$$

It represents the conditional acceptance probability of a pseudo-sample generated under ϑ , given the observed sample. Under the preceding assumptions, joint Borel measurability holds for the mapping

$$(\vartheta, \mathbf{y}^{1:n}) \mapsto w_\tau^n(\vartheta; \mathbf{y}^{1:n}).$$

The corresponding normalization factor is

$$C_\tau^n(\mathbf{Y}^{1:n}) := \int_{\Theta} w_\tau^n(\vartheta; \mathbf{Y}^{1:n}) \pi(d\vartheta). \quad (22)$$

For a given realization of the observed sample, it may vanish when the acceptance probability is zero for π -almost every $\vartheta \in \Theta$; whenever positive, the associated compatibility posterior is defined by

$$\Pi_\tau^n[B; \mathbf{Y}^{1:n}] := \frac{1}{C_\tau^n(\mathbf{Y}^{1:n})} \int_B w_\tau^n(\vartheta; \mathbf{Y}^{1:n}) \pi(d\vartheta), \quad B \in \mathcal{B}(\Theta). \quad (23)$$

In this representation, the posterior distribution (23) may be interpreted as the prior reweighted by the compatibility weights (21). Its large-sample behavior is therefore linked to the asymptotic discrepancy function (15), as made precise by the limiting behavior of these weights.

The limiting discrepancy profile (15) induces a parameter-space counterpart of the finite-sample acceptance mechanism. For the fixed tolerance τ , define the asymptotic compatibility region

$$\Theta_\tau := \{\vartheta \in \Theta \mid \mathcal{T}_\vartheta^\star < \tau\} \in \mathcal{B}(\Theta). \quad (24)$$

In the supercritical regime (19), definition (17) ensures that

$$\pi[\Theta_\tau] > 0. \quad (25)$$

The next result identifies the asymptotic filtering behavior of the compatibility weights (21) relative to Θ_τ .

Theorem 3.1. *Under the standing assumptions of this section, suppose that*

$$\pi[\{\vartheta \in \Theta \mid \mathcal{T}_\vartheta^\star = \tau\}] = 0. \quad (26)$$

Then the compatibility weights converge to the indicator of Θ_τ in the following integrated L^1 sense:

$$\lim_{n \rightarrow \infty} \mathbf{E}_\mathbf{P} \left[\int_{\Theta} |w_\tau^n(\vartheta; \mathbf{Y}^{1:n}) - \mathbb{1}_{\Theta_\tau}(\vartheta)| \pi(d\vartheta) \right] = 0. \quad (27)$$

Proof. Fix $\vartheta \in \Theta$ for which (16) holds, as is the case π -almost everywhere. For each $n \in \mathbb{N}^*$, consider the product probability measure $\mathbf{P} \otimes \mu_\vartheta^n$ on $(\Omega \times \mathcal{Y}^n, \mathcal{A} \otimes \mathcal{B}(\mathcal{Y}^n))$.

By the triangle inequality and symmetry of $\mathcal{T}$, for every $\omega \in \Omega$ and $\mathbf{y}_\vartheta^{1:n} \in \mathcal{Y}^n$,

$$|\Delta_{\vartheta,n}(\mathbf{Y}^{1:n}(\omega), \mathbf{y}_\vartheta^{1:n}) - \mathcal{T}_\vartheta^\star| \leq \mathcal{T}_n^\star(\omega) + \mathcal{T}(\nu_{\vartheta,n}(\mathbf{y}_\vartheta^{1:n}), \nu_\vartheta). \quad (28)$$

Therefore, in view of (13) and (16),

$$\Delta_{\vartheta,n}(\mathbf{Y}^{1:n}(\cdot), \cdot) \longrightarrow \mathcal{T}_\vartheta^\star$$

in probability under the sequence of product measures $\{\mathbf{P} \otimes \mu_\vartheta^n\}_{n \in \mathbb{N}^*}$ as $n \rightarrow \infty$.

Condition (26) allows us to further assume that $\mathcal{T}_\vartheta^\star \neq \tau$, so that the preceding convergence implies

$$\mathbb{1}_{A_{\vartheta,n}(\mathbf{Y}^{1:n}(\cdot); \tau)}(\cdot) \longrightarrow \mathbb{1}_{\Theta_\tau}(\vartheta)$$

in probability and, by boundedness, also in L^1 with respect to the same sequence of product measures.

In conclusion, combining (21) with Tonelli's theorem and dominated convergence,

$$\begin{aligned} \mathbf{E}_\mathbf{P} \left[\int_{\Theta} |w_\tau^n(\vartheta; \mathbf{Y}^{1:n}) - \mathbb{1}_{\Theta_\tau}(\vartheta)| \pi(d\vartheta) \right] &= \int_{\Theta} \mathbf{E}_\mathbf{P} [|w_\tau^n(\vartheta; \mathbf{Y}^{1:n}) - \mathbb{1}_{\Theta_\tau}(\vartheta)|] \pi(d\vartheta) \\ &\leq \int_{\Theta} \left[\int_{\Omega \times \mathcal{Y}^n} |\mathbb{1}_{A_{\vartheta,n}(\mathbf{Y}^{1:n}(\omega); \tau)}(\mathbf{y}_\vartheta^{1:n}) - \mathbb{1}_{\Theta_\tau}(\vartheta)| (\mathbf{P} \otimes \mu_\vartheta^n)(d\omega, d\mathbf{y}_\vartheta^{1:n}) \right] \pi(d\vartheta) \longrightarrow 0 \end{aligned}$$

as $n \rightarrow \infty$, which proves (27) and hence the theorem. $\square$

Theorem 3.1 determines the large-sample behavior of the normalization constants (22) and identifies the asymptotic form of the compatibility posteriors (23) as the conditional prior distribution

$$\pi[\cdot | \Theta_\tau],$$

which is well defined in light of (25). These consequences are made precise in the following corollary.

Corollary 3.1. *Under the assumptions of Theorem 3.1,*

$$\lim_{n \rightarrow \infty} \mathbf{E}_\mathbf{P} [|C_\tau^n(\mathbf{Y}^{1:n}) - \pi[\Theta_\tau]|] = 0, \quad (29)$$

and

$$\lim_{n \rightarrow \infty} \mathbf{P} [C_\tau^n(\mathbf{Y}^{1:n}) = 0] = 0. \quad (30)$$

Moreover, the compatibility posteriors (23) converge in total variation to $\pi[\cdot | \Theta_\tau]$ in $\mathbf{P}$ -probability as $n \rightarrow \infty$, independently of their definition whenever $C_\tau^n(\mathbf{Y}^{1:n}) = 0$.

Proof. By (22) and (27), relation (29) follows immediately and, together with (25), implies (30).

Next, on the event $C_\tau^n(\mathbf{Y}^{1:n}) > 0$, one has, for every $B \in \mathcal{B}(\Theta)$,

$$|\Pi_\tau^n[B; \mathbf{Y}^{1:n}] - \pi[B | \Theta_\tau]| \leq \frac{2}{C_\tau^n(\mathbf{Y}^{1:n})} \int_\Theta |w_\tau^n(\vartheta; \mathbf{Y}^{1:n}) - \mathbb{1}_{\Theta_\tau}(\vartheta)| \pi(d\vartheta).$$

Relation (27), along with (29) and the positivity in (25), shows that the right-hand side converges to zero in $\mathbf{P}$ -probability. Taking the supremum over $B \in \mathcal{B}(\Theta)$ and recalling (30) completes the proof. $\square$

The proof of Theorem 3.1 also provides a direct geometric interpretation of the filtering mechanism. Indeed, by (28), for $\vartheta \in \Theta_\tau$, $\omega \in \Omega$, and $\mathbf{y}_\vartheta^{1:n} \in \mathcal{Y}^n$, the acceptance criterion (20) is met whenever

$$\mathcal{T}_n^\star(\omega) + \mathcal{T}(\nu_{\vartheta,n}(\mathbf{y}_\vartheta^{1:n}), \nu_\vartheta) \leq \tau - \mathcal{T}_\vartheta^\star. \quad (31)$$

The positive compatibility margin

$$\tau_\vartheta^\star := \tau - \mathcal{T}_\vartheta^\star > 0, \quad \vartheta \in \Theta_\tau,$$

thus accommodates both observational and simulated finite-sample fluctuations.

3.3 Compatibility under partial stabilization

The exact filtering results of the previous subsection rely on the asymptotic stabilization of both observed and simulated probabilistic representations, as specified in (13) and (16), respectively. We now replace the latter condition with a weaker assumption, retaining the remaining hypotheses. The compatibility margin in (31) then provides quantitative bounds on asymptotic acceptance probabilities.

Proposition 3.1. *Suppose there exist $B_\tau \in \mathcal{B}(\Theta)$ with $\pi[B_\tau] > 0$ and $\pi[B_\tau \setminus \Theta_\tau] = 0$, and $\delta_\tau \in [0, 1[$ such that, for π -almost every $\vartheta \in B_\tau$, some $\lambda_{\tau,\vartheta} \in]0, 1[$ satisfies*

$$\limsup_{n \rightarrow \infty} \mu_\vartheta^n [\{\mathbf{y}_\vartheta^{1:n} \in \mathcal{Y}^n \mid \mathcal{T}(\nu_{\vartheta,n}(\mathbf{y}_\vartheta^{1:n}), \nu_\vartheta) > \lambda_{\tau,\vartheta} \tau_\vartheta^\star\}] \leq \delta_\tau. \quad (32)$$

Then, for π -almost every $\vartheta \in B_\tau$,

$$\liminf_{n \rightarrow \infty} \mathbf{E}_\mathbf{P} [w_\tau^n(\vartheta; \mathbf{Y}^{1:n})] \geq 1 - \delta_\tau. \quad (33)$$

Proof. Fix $\vartheta \in B_\tau$ for which (32) holds. By (31), for any $\omega \in \Omega$ and $\mathbf{y}_\vartheta^{1:n} \in \mathcal{Y}^n$,

$$\begin{cases} \mathcal{T}_n^\star(\omega) \leq (1 - \lambda_{\tau,\vartheta}) \tau_\vartheta^\star \\ \mathcal{T}(\nu_{\vartheta,n}(\mathbf{y}_\vartheta^{1:n}), \nu_\vartheta) \leq \lambda_{\tau,\vartheta} \tau_\vartheta^\star \end{cases} \implies \mathbf{y}_\vartheta^{1:n} \in A_{\vartheta,n}(\mathbf{Y}^{1:n}(\omega); \tau).$$

Hence, by (21) and the product structure,

$$\mathbf{E}_\mathbf{P} [w_\tau^n(\vartheta; \mathbf{Y}^{1:n})] \geq \mathbf{P} [\mathcal{T}_n^\star \leq (1 - \lambda_{\tau,\vartheta}) \tau_\vartheta^\star] \mu_\vartheta^n [\{\mathbf{y}_\vartheta^{1:n} \in \mathcal{Y}^n \mid \mathcal{T}(\nu_{\vartheta,n}(\mathbf{y}_\vartheta^{1:n}), \nu_\vartheta) \leq \lambda_{\tau,\vartheta} \tau_\vartheta^\star\}].$$

Since $(1 - \lambda_{\tau,\vartheta}) \tau_\vartheta^\star > 0$, relations (13) and (32) imply that the first factor on the right-hand side converges to one and the lower limit of the second is at least $1 - \delta_\tau$, as claimed in (33). $\square$

An integrated version of Proposition 3.1 follows directly from Tonelli's theorem and Fatou's lemma.

Corollary 3.2. *Under the assumptions of Proposition 3.1,*

$$\liminf_{n \rightarrow \infty} \mathbb{E}_{\mathbf{P}} \left[\int_{B_\tau} w_\tau^n(\vartheta; Y^{1:n}) \pi(d\vartheta) \right] \geq (1 - \delta_\tau) \pi[B_\tau]. \quad (34)$$

Corollary 3.2 shows that exact stabilization of the simulated representations is not necessary for compatible prior mass to persist asymptotically. The residual level δ_τ quantifies the admissible loss, while B_τ identifies the portion of the parameter space on which this control is retained.

We finally relate these persistence bounds to the geometry of the asymptotic discrepancy function (15). We assume that the model-wide compatibility threshold (18) is attained by at least one parameter, so that

$$\varepsilon_\Theta^* = \min_{\vartheta \in \Theta} \mathcal{T}_\vartheta^*, \quad (35)$$

and we then introduce the nonempty measurable set

$$\Theta^* := \{ \vartheta^* \in \Theta \mid \mathcal{T}_{\vartheta^*}^* = \varepsilon_\Theta^* \}, \quad (36)$$

the distance function to Θ^* defined, for every $\vartheta \in \Theta$, by

$$d_{\Theta^*}(\vartheta) := \inf_{\vartheta^* \in \Theta^*} d_\Theta(\vartheta, \vartheta^*),$$

and, for every $r \in]0, \infty[$, the open r -neighborhood of Θ^* in Θ ,

$$\Theta_r^* := \{ \vartheta \in \Theta \mid d_{\Theta^*}(\vartheta) < r \}. \quad (37)$$

Proposition 3.2. *Suppose there exist $r^* \in]0, \infty[$ and two non-decreasing functions*

$$\psi_-, \psi_+ : [0, r^*] \longrightarrow \mathbb{R}_+$$

such that, for π -almost every $\vartheta \in \Theta_{r^}^*$,*

$$\psi_-(d_{\Theta^*}(\vartheta)) \leq \mathcal{T}_\vartheta^* - \varepsilon_\Theta^* \leq \psi_+(d_{\Theta^*}(\vartheta)). \quad (38)$$

Then, for any $r_-, r_+ \in]0, r^[$ satisfying*

$$\psi_+(r_-) < \tau - \varepsilon_\Theta^* \leq \psi_-(r_+), \quad (39)$$

one has

$$\pi[\Theta_{r_-}^* \setminus \Theta_\tau] = \pi[(\Theta_\tau \cap \Theta_{r_+}^*) \setminus \Theta_{r_+}^*] = 0. \quad (40)$$

Proof. For π -almost every $\vartheta \in \Theta_{r_-}^* \subseteq \Theta_{r^*}^*$ (see (37)), in virtue of the upper bound in (38), monotonicity of ψ_+ , and the first inequality in (39), we obtain

$$\mathcal{T}_\vartheta^* \leq \varepsilon_\Theta^* + \psi_+(d_{\Theta^*}(\vartheta)) \leq \varepsilon_\Theta^* + \psi_+(r_-) < \tau,$$

hence $\vartheta \in \Theta_\tau$ by (24). Similarly, for π -almost every $\vartheta \in \Theta_{r^*}^* \setminus \Theta_{r_+}^*$, the lower bound in (38), definition (37), monotonicity of ψ_- , and the second inequality in (39) give

$$\mathcal{T}_\vartheta^* \geq \varepsilon_\Theta^* + \psi_-(d_{\Theta^*}(\vartheta)) \geq \varepsilon_\Theta^* + \psi_-(r_+) \geq \tau,$$

which implies $\vartheta \notin \Theta_\tau$. The two conclusions together establish (40). $\square$

Proposition 3.2 offers a two-sided geometric description of the asymptotic compatibility region Θ_τ around Θ^* . The function ψ_+ identifies neighborhoods that remain asymptotically compatible, while ψ_- bounds the local extent of Θ_τ away from the set of model-wide discrepancy minimizers.

Remark 3.3. Whenever ψ_- and ψ_+ are continuous and strictly increasing, the inner and outer geometry of the compatibility region is governed, respectively, by the inverse scales

$$\psi_+^{-1}(\tau - \varepsilon_\Theta^\star), \psi_-^{-1}(\tau - \varepsilon_\Theta^\star),$$

provided that both are well defined and belong to $]0, r^\star[$. Thus, flatter discrepancy profiles may produce compatibility regions extending farther from $\Theta^\star$, while faster growth leads to tighter localization.

Under polynomial discrepancy growth, the same argument makes the compatibility scales explicit.

Corollary 3.3. *Let $r^\star, q, c_-, c_+ \in]0, \infty[$ with $c_- \leq c_+$ be such that, for π -almost every $\vartheta \in \Theta_{r^\star}^\star$,*

$$c_- d_{\Theta^\star}^q(\vartheta) \leq \mathcal{T}_\vartheta^\star - \varepsilon_\Theta^\star \leq c_+ d_{\Theta^\star}^q(\vartheta).$$

Assuming that

$$\left(\frac{\tau - \varepsilon_\Theta^\star}{c_-} \right)^{1/q} < r^\star,$$

the following two inclusions hold up to π -null sets:

$$\Theta_{\left(\frac{\tau - \varepsilon_\Theta^\star}{c_+}\right)^{1/q}}^\star \subseteq \Theta_\tau, \quad \Theta_\tau \cap \Theta_{r^\star}^\star \subseteq \Theta_{\left(\frac{\tau - \varepsilon_\Theta^\star}{c_-}\right)^{1/q}}^\star.$$

The inner localization in Proposition 3.2 can also be coupled with Proposition 3.1 and Corollary 3.2: if $\pi[\Theta_{r_-}^\star] > 0$ and condition (32) holds with a uniform δ_τ for π -almost every $\vartheta \in \Theta_{r_-}^\star$, we may take

$$B_\tau = \Theta_{r_-}^\star.$$

Consequently, inequality (34) specializes to

$$\liminf_{n \rightarrow \infty} \mathbf{E}_\mathbf{P} \left[\int_{\Theta_{r_-}^\star} w_\tau^n(\vartheta; \mathbf{Y}^{1:n}) \pi(d\vartheta) \right] \geq (1 - \delta_\tau) \pi[\Theta_{r_-}^\star].$$

Observe further that, in the polynomial case of Corollary 3.3, one may choose any $r_- > 0$ satisfying

$$r_- < \left(\frac{\tau - \varepsilon_\Theta^\star}{c_+} \right)^{1/q}.$$

Taken together, the results of this section distinguish two complementary large-sample regimes. Under full stabilization of the observed and simulated probabilistic representations, the posterior converges to the prior filtered by Θ_τ . Under partial stabilization, exact filtering may fail, while positive prior-averaged acceptance can persist quantitatively within neighborhoods controlled by the local growth of the asymptotic discrepancy. This analysis provides the foundation for the next section.

4 Variational and transport structures in ABC

We proceed from the probabilistic and asymptotic analysis of compatibility-based inference to its variational and geometric structures. We first employ entropy-regularized minimization principles to characterize posterior distributions and establish the stability of the associated variational problems as the compatibility threshold vanishes. We then specialize the asymptotic discrepancy to a Wasserstein distance and develop the corresponding geometric representation on the space of probability measures.

The two perspectives are connected through an exact reduction from the parameter space to the space of model laws, preserving the minimum value and the correspondence between minimizers.

4.1 A compatibility-induced variational principle

Throughout the subsection, let $\varepsilon > 0$ be given, and let $w_\varepsilon: \Theta \rightarrow \mathbb{R}_+$ be a compatibility weight as introduced in Subsection 2.1. From a variational perspective, rather than viewing w_ε solely as quantifying the compatibility of each parameter value $\vartheta \in \Theta$ with the observed sample $\mathbf{y}^{1:n} \in \mathcal{Y}^n$, we regard it as defining a data-dependent loss structure on the parameter space.

Henceforth, we restrict our attention to compatibility weights satisfying, π -a.s.,

$$w_\varepsilon \leq 1. \quad (41)$$

Up to a modification on a π -null set, we may assume that (41) holds everywhere on Θ , and introduce the corresponding extended-valued compatibility loss ℓ_ε by setting, for any $\vartheta \in \Theta$,

$$\ell_\varepsilon(\vartheta) := \begin{cases} -\log w_\varepsilon(\vartheta), & \text{if } w_\varepsilon(\vartheta) > 0, \\ \infty, & \text{otherwise.} \end{cases} \quad (42)$$

This logarithmic transformation converts the multiplicative compatibility scale into an additive one, playing a role analogous to that of the negative log-likelihood in classical statistical inference. Larger values of w_ε correspond to smaller values of ℓ_ε , so that minimizing the compatibility loss is equivalent to maximizing compatibility with the observed sample. Under the present assumptions, the induced compatibility loss is nonnegative, consistent with the classical ABC setting (see also Subsection 2.2). This additive formulation is particularly well suited for variational analysis, as it naturally gives rise to entropy-regularized problems in approximate Bayesian inference.

A further connection arises with loss-based forms of generalized Bayesian updating; see, e.g., [5]. The compatibility loss (42) plays the role of an empirical loss, while the compatibility weights act as likelihood surrogates. The resulting inferential structure is induced by the discrepancy geometry relating observed and simulated data, without requiring explicit evaluation of the observation likelihood.

At the level of posterior candidates, this pointwise loss induces the extended-valued functional

$$\varrho \mapsto \int_{\Theta} \ell_\varepsilon(\vartheta) \varrho(d\vartheta), \quad \varrho \in \mathcal{P}(\Theta), \quad (43)$$

which represents the compatibility loss averaged with respect to the candidate posterior distribution ϱ . Thus, posterior candidates are no longer evaluated only through the compatibility of individual parameter values, but through the aggregate loss associated with their distribution over the parameter space.

This formulation extends to a general variational framework suited for the subsequent developments. It is based on the interplay between a data-fitting functional

$$\mathcal{R}_\varepsilon: \mathcal{P}(\Theta) \longrightarrow [0, \infty]$$

and a regularization functional

$$I(\cdot | \pi): \mathcal{P}(\Theta) \longrightarrow [0, \infty],$$

with the former representing the compatibility component associated with the observed sample $\mathbf{y}^{1:n}$, and the latter acting as a prior-based informational regularization term. This yields the variational problem

$$\inf_{\varrho \in \mathcal{P}(\Theta)} \{ \mathcal{R}_\varepsilon(\varrho) + I(\varrho | \pi) \}. \quad (44)$$

In the canonical compatibility-loss setting introduced above, the functional $\mathcal{R}_\varepsilon$ reduces precisely to (43). In this case, it admits a natural interpretation as a compatibility-induced risk functional, thereby motivating its notation. The regularization term then acts to prevent the minimization from collapsing into a purely compatibility-driven concentration on the most compatible regions of the parameter space, while preserving the role of the prior as an informational reference measure.

Among the possible choices for the functional $I(\cdot | \pi)$, the Kullback–Leibler divergence, also known as relative entropy, emerges as a particularly natural candidate. It is defined on $\mathcal{P}(\Theta)$ by

$$\text{KL}(\varrho | \pi) := \begin{cases} \int_{\Theta} \log \left[\frac{d\varrho}{d\pi}(\vartheta) \right] \varrho(d\vartheta), & \text{if } \varrho \ll \pi, \\ \infty, & \text{otherwise.} \end{cases} \quad (45)$$

The relative entropy quantifies the informational divergence of the posterior candidate ϱ from the prior distribution π . Within this framework, it acts as an informational regularization term, discouraging excessive departures from the prior while balancing the influence of the compatibility component.

Within this compatibility-loss setting, and taking the Kullback–Leibler divergence as the regularization functional, the variational problem (44) specializes to

$$\inf_{\varrho \in \mathcal{P}(\Theta)} \left\{ \int_{\Theta} \ell_{\varepsilon}(\vartheta) \varrho(d\vartheta) + \text{KL}(\varrho \mid \pi) \right\}. \quad (46)$$

This is an entropy-regularized risk minimization problem over $\mathcal{P}(\Theta)$, balancing compatibility with the observed data against informational divergence from the prior.

The compatibility-weight posterior distribution $\pi_{\varepsilon} \in \mathcal{P}(\Theta)$ introduced in (4) admits a variational characterization in terms of the objective functional $\mathcal{F}_{\varepsilon}$ on $\mathcal{P}(\Theta)$ given by

$$\mathcal{F}_{\varepsilon}(\varrho) := \int_{\Theta} \ell_{\varepsilon}(\vartheta) \varrho(d\vartheta) + \text{KL}(\varrho \mid \pi). \quad (47)$$

This representation is a compatibility-based instance of the classical Gibbs–Donsker–Varadhan principle; see, e.g., [10, 9]. Although standard, it is included for completeness, as it provides the fundamental link between compatibility-weight posteriors and entropy-regularized risk minimization.

Proposition 4.1. *Under the standing assumptions of this subsection, the probability measure π_{ε} , defined as in (4), is the unique minimizer of $\mathcal{F}_{\varepsilon}$ on $\mathcal{P}(\Theta)$, with minimum value*

$$\min_{\varrho \in \mathcal{P}(\Theta)} \mathcal{F}_{\varepsilon}(\varrho) \equiv \mathcal{F}_{\varepsilon}(\pi_{\varepsilon}) = -\log \int_{\Theta} w_{\varepsilon}(\vartheta) \pi(d\vartheta).$$

Proof. Let $C_{\varepsilon} \in]0, 1]$ denote the normalization constant associated with π_{ε} , that is,

$$C_{\varepsilon} := \int_{\Theta} w_{\varepsilon}(\vartheta) \pi(d\vartheta). \quad (48)$$

By construction, $\pi_{\varepsilon} \ll \pi$, with

$$\frac{d\pi_{\varepsilon}}{d\pi} = \frac{w_{\varepsilon}}{C_{\varepsilon}}.$$

Hence, for π -almost every $\vartheta \in \Theta$ such that $w_{\varepsilon}(\vartheta) > 0$,

$$\log w_{\varepsilon}(\vartheta) = \log C_{\varepsilon} + \log \frac{d\pi_{\varepsilon}}{d\pi}(\vartheta).$$

Now, fix $\varrho \in \mathcal{P}(\Theta)$. The preceding identities and the extended-valued definitions (42) and (45) give the relation below, whether $\varrho \ll \pi_{\varepsilon}$ or $\varrho \not\ll \pi_{\varepsilon}$:

$$\mathcal{F}_{\varepsilon}(\varrho) = -\log C_{\varepsilon} + \text{KL}(\varrho \mid \pi_{\varepsilon}). \quad (49)$$

The non-negativity of the Kullback–Leibler divergence in (49), together with the fact that it vanishes if and only if $\varrho = \pi_{\varepsilon}$, yields the stated characterization. $\square$

Proposition 4.1 shows that the posterior distribution π_{ε} is not merely the normalization of compatibility weights, but the unique minimizer of an entropy-regularized compatibility functional.

Posterior updating may thus be interpreted as a balance between compatibility with the observed data and informational coherence with the prior distribution. The normalization constant (48) plays the role of a partition function, while its negative logarithm represents the minimum value of this functional.

At this stage, a natural question concerns the stability of the variational characterization above under the vanishing-threshold behavior of compatibility weights identified in Proposition 2.1. The next result confirms that this stability persists for the limiting posterior.

Theorem 4.1. *Under the standing assumptions of this subsection and of Proposition 2.1, define*

$$\tilde{\mathcal{F}}_\varepsilon(\varrho) := \mathcal{F}_\varepsilon(\varrho) + \log \kappa_\varepsilon, \quad \varrho \in \mathcal{P}(\Theta), \quad (50)$$

with $\mathcal{F}_\varepsilon$ as in (47), and

$$\bar{\mathcal{F}}(\varrho) := -\log \int_{\Theta} \bar{w}(\vartheta) \pi(d\vartheta) + \text{KL}(\varrho | \bar{\pi}), \quad \varrho \in \mathcal{P}(\Theta), \quad (51)$$

where $\bar{\pi}$ is given by (6). Then π_ε is the unique minimizer of $\tilde{\mathcal{F}}_\varepsilon$ on $\mathcal{P}(\Theta)$, with minimum value

$$\min_{\varrho \in \mathcal{P}(\Theta)} \tilde{\mathcal{F}}_\varepsilon(\varrho) \equiv \tilde{\mathcal{F}}_\varepsilon(\pi_\varepsilon) = -\log \int_{\Theta} \frac{1}{\kappa_\varepsilon} w_\varepsilon(\vartheta) \pi(d\vartheta). \quad (52)$$

Likewise, the probability measure $\bar{\pi}$ is the unique minimizer of $\bar{\mathcal{F}}$ on $\mathcal{P}(\Theta)$, and

$$\min_{\varrho \in \mathcal{P}(\Theta)} \bar{\mathcal{F}}(\varrho) \equiv \bar{\mathcal{F}}(\bar{\pi}) = -\log \int_{\Theta} \bar{w}(\vartheta) \pi(d\vartheta). \quad (53)$$

Furthermore,

$$\lim_{\varepsilon \downarrow 0} \tilde{\mathcal{F}}_\varepsilon(\pi_\varepsilon) = \bar{\mathcal{F}}(\bar{\pi}). \quad (54)$$

Finally, if $\{\varrho_\varepsilon\}_{\varepsilon > 0}$ is any family in $\mathcal{P}(\Theta)$ of asymptotic minimizers, that is,

$$\lim_{\varepsilon \downarrow 0} \left[\tilde{\mathcal{F}}_\varepsilon(\varrho_\varepsilon) - \tilde{\mathcal{F}}_\varepsilon(\pi_\varepsilon) \right] = 0, \quad (55)$$

then this family converges in total variation to $\bar{\pi}$.

Proof. The characterization of π_ε as the unique minimizer of $\tilde{\mathcal{F}}_\varepsilon$, together with identity (52), follows directly from Proposition 4.1. The corresponding statement for $\bar{\mathcal{F}}$ is immediate from definition (51). Identity (54) is then obtained by combining (52) and (53) with the $L^1(\pi)$ -convergence (5) as $\varepsilon \downarrow 0$.

It remains to demonstrate the stability of asymptotic minimizers $\{\varrho_\varepsilon\}_{\varepsilon > 0}$. By (50) and (49), one has

$$\tilde{\mathcal{F}}_\varepsilon(\varrho_\varepsilon) - \tilde{\mathcal{F}}_\varepsilon(\pi_\varepsilon) = \text{KL}(\varrho_\varepsilon | \pi_\varepsilon),$$

and, in particular, condition (55) may be rewritten as

$$\lim_{\varepsilon \downarrow 0} \text{KL}(\varrho_\varepsilon | \pi_\varepsilon) = 0. \quad (56)$$

On the other hand, Pinsker's inequality gives

$$\sup_{B \in \mathcal{B}(\Theta)} |\varrho_\varepsilon[B] - \pi_\varepsilon[B]| \leq \sqrt{\frac{1}{2} \text{KL}(\varrho_\varepsilon | \pi_\varepsilon)}, \quad (57)$$

which, combined with (56), shows that the left-hand side in (57) also vanishes as $\varepsilon \downarrow 0$.

In conclusion, since Proposition 2.1 ensures that $\{\pi_\varepsilon\}_\varepsilon$ converges to $\bar{\pi}$ in total variation, an application of the triangle inequality implies the same convergence for $\{\varrho_\varepsilon\}_{\varepsilon > 0}$. $\square$

Theorem 4.1 carries implications that extend beyond the posterior convergence established in Proposition 2.1. Indeed, the variational structure persists in the asymptotic regime: the limiting profile induces a variational problem governed by the same principle as its finite-threshold counterparts.

From a risk-based perspective, the theorem shows that every family of posterior candidates whose excess entropy-regularized compatibility risk tends to zero converges to $\bar{\pi}$ in total variation.

Remark 4.1. The preceding stability result suggests a connection with variational convergence theory, in particular with Γ -convergence; see, e.g., [8]. Indeed, convergence of minimum values and stability of asymptotic minimizers are characteristic consequences of suitable variational convergence principles.

A rigorous treatment of this connection would require the development of an explicit Γ -convergence theory, a perspective that we leave for future investigation.

The variational stability established in Theorem 4.1 reveals an underlying principle that is not specific to the entropy-based setting considered above. In the present framework, relative entropy defines the informational geometry associated with the prior, while alternative regularizers may induce different notions of statistical complexity, regularity, or proximity.

The asymptotic regime developed in Section 3 provides a transparent realization of this broader perspective. Indeed, whenever the limiting discrepancy profile (15) is available,

$$\vartheta \mapsto \mathcal{T}_\vartheta^\star, \quad \vartheta \in \Theta,$$

it induces a canonical asymptotic notion of compatibility on the parameter space, since, for each $\vartheta \in \Theta$, the value $\mathcal{T}_\vartheta^\star$ measures the asymptotic discrepancy between the probabilistic structure generated under ϑ and the limiting observational distribution associated with the data. Hence, smaller values correspond to statistical models exhibiting a higher degree of asymptotic compatibility with the observed sample.

This naturally leads to the asymptotic compatibility functional

$$\mathcal{R}_\varphi^\star(\varrho) := \int_{\Theta} \varphi(\mathcal{T}_\vartheta^\star - \varepsilon_\Theta^\star) \varrho(d\vartheta), \quad \varrho \in \mathcal{P}(\Theta), \quad \varrho \ll \pi \quad (58)$$

(understood in the extended sense) associated with a prescribed strictly increasing penalty function

$$\varphi: \mathbb{R}_+ \longrightarrow \mathbb{R}_+,$$

continuous at zero and satisfying $\varphi(0) = 0$, where $\varepsilon_\Theta^\star$ is the model-wide compatibility threshold (18).

The profile $\vartheta \mapsto \mathcal{T}_\vartheta^\star$ determines the intrinsic asymptotic compatibility structure on Θ , while the transformation φ specifies how deviations from the optimal compatibility level are quantified. The resulting functional $\mathcal{R}_\varphi^\star$ evaluates the average penalized asymptotic incompatibility of posterior candidates, separating the asymptotic discrepancy profile from the penalization mechanism.

Remark 4.2. The functional $\mathcal{R}_\varphi^\star$ extends the asymptotic compatibility structure from Θ to probability measures absolutely continuous with respect to π , quantifying their penalized incompatibility.

This interpretation is consistent with the distinction between the prior-essential and model-wide compatibility thresholds (17) and (18), respectively. Indeed, for every $\varrho \in \mathcal{P}(\Theta)$ with $\varrho \ll \pi$, one has

$$\mathcal{R}_\varphi^\star(\varrho) \geq \varphi(\varepsilon_\pi^\star - \varepsilon_\Theta^\star),$$

and, in particular, if $\varepsilon_\pi^\star = \varepsilon_\Theta^\star$, then

$$\inf_{\substack{\varrho \in \mathcal{P}(\Theta) \\ \varrho \ll \pi}} \mathcal{R}_\varphi^\star(\varrho) = 0.$$

Whenever (35) holds and the associated minimal set $\Theta^\star$ defined in (36) satisfies $\pi[\Theta^\star] > 0$, the infimum is attained precisely by the measures $\varrho \ll \pi$ satisfying $\varrho[\Theta^\star] = 1$.

The functional (58) thus provides a variational counterpart to the asymptotic compatibility structure and the associated persistence properties developed in Section 3.

Among the possible geometric extensions, an optimal-transport approach constitutes a natural specialization of the framework introduced so far.

As anticipated in Section 3, transport-based choices of the discrepancy map (9) induce compatibility mechanisms in Approximate Bayesian Computation. The same geometric principles also allow posterior candidates to be represented through probability distributions on the space of model laws.

These considerations motivate the formulation introduced in the next subsection.

4.2 Optimal transport structures

In this subsection, compatibility is characterized via transport discrepancies on the data space, while the associated variational structure is represented through the transport geometry of the induced model laws. The analysis is carried out within the theory of optimal transport; see, e.g., the foundational contribution [22], the classical monographs [39, 40], and the more recent treatments [2, 36, 35].

The observation space $(\mathcal{Y}, d_{\mathcal{Y}})$ is assumed to be Polish. With separability already ensured, the only additional hypothesis is completeness. This is the case, for instance, when $\mathcal{Y}$ is a closed subset of $\mathbb{R}^{m_{\mathcal{Y}}}$ endowed with the induced Euclidean metric. Besides encompassing most statistical models encountered in practice, this provides the regularity needed for the transport-theoretic constructions.

We restrict attention to the classical Wasserstein setting generated by the ground metric $d_{\mathcal{Y}}$, although several of the constructions below admit extensions to more general transport costs.

To this end, we fix once and for all an exponent

$$p \in [1, \infty[.$$

Let $\mathcal{P}_p(\mathcal{Y})$ denote the subclass of $\mathcal{P}(\mathcal{Y})$ consisting of all probability measures η on $\mathcal{Y}$ admitting a finite p -th moment; i.e., for some (equivalently, every) $y_0 \in \mathcal{Y}$,

$$\int_{\mathcal{Y}} d_{\mathcal{Y}}^p(y, y_0) \eta(dy) < \infty.$$

We equip $\mathcal{P}_p(\mathcal{Y})$ with the p -Wasserstein distance associated with $d_{\mathcal{Y}}$, defined by

$$W_p(\mu, \nu) := \left[\inf_{\gamma \in \Pi(\mu, \nu)} \int_{\mathcal{Y} \times \mathcal{Y}} d_{\mathcal{Y}}^p(y, z) \gamma(dy, dz) \right]^{1/p}, \quad \mu, \nu \in \mathcal{P}_p(\mathcal{Y}),$$

where $\Pi(\mu, \nu)$ denotes the set of all couplings of μ and ν , that is, all probability measures on $\mathcal{Y} \times \mathcal{Y}$ whose first and second marginals are μ and ν , respectively.

The above definition coincides with the Kantorovich formulation of the optimal transport problem corresponding to the cost function $d_{\mathcal{Y}}^p$. Under the present assumptions, this problem admits a minimizer, namely an optimal transport plan. Moreover, the Wasserstein space $(\mathcal{P}_p(\mathcal{Y}), W_p)$ is again Polish.

Accordingly, the framework introduced in Subsection 3.1 is henceforth understood upon replacing $\mathcal{P}(\mathcal{Y})$ with $\mathcal{P}_p(\mathcal{Y})$. Specifically, for every $\mu, \nu \in \mathcal{P}_p(\mathcal{Y})$, we set

$$\mathcal{T}(\mu, \nu) := W_p(\mu, \nu).$$

The statistical model associates with every parameter value $\vartheta \in \Theta$ the corresponding law $\nu_{\vartheta} \in \mathcal{P}_p(\mathcal{Y})$, introduced in (14). The family $\{\nu_{\vartheta}\}_{\vartheta \in \Theta}$ provides a representation of the model class in $\mathcal{P}_p(\mathcal{Y})$, while the limiting measure $\mu^* \in \mathcal{P}_p(\mathcal{Y})$, specified in (11), encodes the observational information.

In particular, the asymptotic compatibility profile (15) is given, for every $\vartheta \in \Theta$, by

$$\mathcal{T}_{\vartheta}^* = W_p(\mu^*, \nu_{\vartheta}). \quad (59)$$

Under the attainment assumption (35), the model-wide compatibility threshold (18) becomes

$$\varepsilon_{\Theta}^* = \min_{\vartheta \in \Theta} W_p(\mu^*, \nu_{\vartheta}).$$

The asymptotic compatibility functional (58) then takes the explicit Wasserstein form

$$\mathcal{R}_{\varphi, p}^*(\varrho) := \int_{\Theta} \varphi(W_p(\mu^*, \nu_{\vartheta}) - \varepsilon_{\Theta}^*) \varrho(d\vartheta), \quad \varrho \in \mathcal{P}(\Theta), \quad \varrho \ll \pi, \quad (60)$$

where $\varphi: \mathbb{R}_+ \rightarrow \mathbb{R}_+$ is the strictly increasing penalty function continuous at zero with $\varphi(0) = 0$.

Under (35), Remark 4.2 implies that, for every $\varrho \in \mathcal{P}(\Theta)$ with $\varrho \ll \pi$,

$$\mathcal{R}_{\varphi, p}^*(\varrho) = 0 \iff \varrho[\Theta^*] = 1,$$

where Θ^* is the asymptotically optimal set introduced in (36).

Altogether, these observations show that asymptotic compatibility depends on the parameters only through the corresponding model laws. It is therefore natural to represent asymptotic compatibility directly on the Wasserstein space $\mathcal{P}_p(\mathcal{Y})$ of model laws.

To formalize this idea, we assume that the parameter-to-distribution map $\Phi: \Theta \rightarrow \mathcal{P}_p(\mathcal{Y})$ defined by

$$\Phi(\vartheta) = \nu_{\vartheta}, \quad \vartheta \in \Theta, \quad (61)$$

is Borel measurable. In this way, every $\varrho \in \mathcal{P}(\Theta)$, $\varrho \ll \pi$, induces the push-forward (or image) measure

$$\Lambda_{\varrho} := \Phi_{\#} \varrho \in \mathcal{P}(\mathcal{P}_p(\mathcal{Y})), \quad (62)$$

which represents the distribution of the random probability measure $\Phi(\vartheta) = \nu_{\vartheta}$ when the parameter ϑ is distributed according to ϱ . Indeed, by definition,

$$\Phi_{\#} \varrho[\mathcal{E}] = \varrho[\Phi^{-1}(\mathcal{E})], \quad \mathcal{E} \in \mathcal{B}(\mathcal{P}_p(\mathcal{Y})).$$

Moreover, it immediately follows that $\Lambda_{\varrho} \ll \Lambda_{\pi}$, so that

$$W_p(\mu^{\star}, \nu) \geq \varepsilon_{\Theta}^{\star} \quad \Lambda_{\varrho}\text{-a.s.} \quad (63)$$

Applying the standard integration formula for image measures to (60), we obtain

$$\mathcal{R}_{\varphi,p}^{\star}(\varrho) = \int_{\mathcal{P}_p(\mathcal{Y})} \varphi(W_p(\mu^{\star}, \nu) - \varepsilon_{\Theta}^{\star}) \Lambda_{\varrho}(d\nu), \quad \varrho \in \mathcal{P}(\Theta), \varrho \ll \pi. \quad (64)$$

Hence, the asymptotic compatibility functional depends on posterior candidates only through their distributions of model laws. Statistical information is carried by the transport discrepancy from the observational law $\mu^{\star}$, while ϱ enters only through the push-forward measure Λ_{ϱ} .

In particular, any two probability measures $\varrho, \tilde{\varrho} \in \mathcal{P}(\Theta)$ with $\varrho, \tilde{\varrho} \ll \pi$ are indistinguishable with respect to asymptotic transport compatibility whenever they share the same induced law; namely,

$$\Lambda_{\varrho} = \Lambda_{\tilde{\varrho}} \implies \mathcal{R}_{\varphi,p}^{\star}(\varrho) = \mathcal{R}_{\varphi,p}^{\star}(\tilde{\varrho}).$$

The measure (62) thus serves as a natural transport-geometric representation of each posterior candidate.

The preceding representation reveals a direct connection between the entropy-regularized variational structure developed in Subsection 4.1 and the induced distributions of model laws. The next result makes this connection precise by establishing an exact reduction of relative entropy (45) under the map Φ .

Proposition 4.2. *Under the standing assumptions of this subsection, for every $\Lambda \in \mathcal{P}(\mathcal{P}_p(\mathcal{Y}))$,*

$$\inf_{\substack{\varrho \in \mathcal{P}(\Theta) \\ \Phi_{\#} \varrho = \Lambda}} \text{KL}(\varrho | \pi) = \text{KL}(\Lambda | \Lambda_{\pi}), \quad (65)$$

where the divergence on the right-hand side is understood between probability measures on $\mathcal{P}_p(\mathcal{Y})$. Whenever $\Lambda \ll \Lambda_{\pi}$, the infimum is attained by the measure $\varrho_{\Lambda} \in \mathcal{P}(\Theta)$ satisfying $\varrho_{\Lambda} \ll \pi$ and

$$\frac{d\varrho_{\Lambda}}{d\pi} = \frac{d\Lambda}{d\Lambda_{\pi}} \circ \Phi \quad \pi\text{-a.s.} \quad (66)$$

Proof. Let $\Lambda \in \mathcal{P}(\mathcal{P}_p(\mathcal{Y}))$. For every $\varrho \in \mathcal{P}(\Theta)$ such that $\Phi_{\#} \varrho = \Lambda$, i.e., $\Lambda_{\varrho} = \Lambda$ in the notation of (62), the data-processing inequality for relative entropy yields

$$\text{KL}(\Lambda | \Lambda_{\pi}) \leq \text{KL}(\varrho | \pi)$$

(see, e.g., [11]). Minimizing over these measures establishes the “ $\geq$ ” inequality in (65).

For the reverse inequality, if $\Lambda \ll \Lambda_{\pi}$, then $\text{KL}(\Lambda | \Lambda_{\pi}) < \infty$, and the same bound in turn implies that

$$\text{KL}(\varrho | \pi) = \infty$$

for all $\varrho \in \mathcal{P}(\Theta)$ with $\Phi_{\#} \varrho = \Lambda$. Thus, (65) follows directly in this case.

Suppose now that $\Lambda \ll \Lambda_{\pi}$. By the integration formula for image measures, it is readily seen that (66) defines a probability measure $\varrho_{\Lambda} \in \mathcal{P}(\Theta)$; moreover, $\Phi_{\#} \varrho_{\Lambda} = \Lambda$ since, for every $\mathcal{E} \in \mathcal{B}(\mathcal{P}_p(\mathcal{Y}))$,

$$\Phi_{\#} \varrho_{\Lambda}[\mathcal{E}] = \int_{\Theta} \mathbb{1}_{\mathcal{E}}(\Phi(\vartheta)) \frac{d\Lambda}{d\Lambda_{\pi}}(\Phi(\vartheta)) \pi(d\vartheta) = \int_{\mathcal{E}} \frac{d\Lambda}{d\Lambda_{\pi}}(\nu) \Lambda_{\pi}(d\nu) = \Lambda[\mathcal{E}].$$

Finally, a simple calculation based on (66) and this identity gives

$$\text{KL}(\varrho_{\Lambda} | \pi) = \text{KL}(\Lambda | \Lambda_{\pi}),$$

which concludes the proof. $\square$

In conjunction with the push-forward representation (64), Proposition 4.2 provides an exact reduction of an entropy-regularized asymptotic compatibility problem, analogous to (46). For this purpose, set

$$c_{\varphi,p}^*(\nu) := \varphi\left([W_p(\mu^*, \nu) - \varepsilon_\Theta^*]_+\right), \quad \nu \in \mathcal{P}_p(\mathcal{Y}), \quad (67)$$

where $[\cdot]_+$ denotes the positive-part operator on $\mathbb{R}$.

Corollary 4.1. *Under the standing assumptions of this subsection,*

$$\inf_{\substack{\varrho \in \mathcal{P}(\Theta) \\ \varrho \ll \pi}} \{\mathcal{R}_{\varphi,p}^*(\varrho) + \text{KL}(\varrho \mid \pi)\} = \inf_{\Lambda \in \mathcal{P}(\mathcal{P}_p(\mathcal{Y}))} \left\{ \int_{\mathcal{P}_p(\mathcal{Y})} c_{\varphi,p}^*(\nu) \Lambda(d\nu) + \text{KL}(\Lambda \mid \Lambda_\pi) \right\}. \quad (68)$$

The respective problems admit unique minimizers $\varrho_{\varphi,p}^* \ll \pi$ and $\Lambda_{\varphi,p}^* = \Phi_\# \varrho_{\varphi,p}^* \ll \Lambda_\pi$ with

$$\frac{d\Lambda_{\varphi,p}^*}{d\Lambda_\pi} \propto e^{-c_{\varphi,p}^*} \quad \Lambda_\pi\text{-a.s.}, \quad \frac{d\varrho_{\varphi,p}^*}{d\pi} = \frac{d\Lambda_{\varphi,p}^*}{d\Lambda_\pi} \circ \Phi \quad \pi\text{-a.s.} \quad (69)$$

Proof. For every $\varrho \in \mathcal{P}(\Theta)$ with $\varrho \ll \pi$, by (64) and (67),

$$\mathcal{R}_{\varphi,p}^*(\varrho) = \int_{\mathcal{P}_p(\mathcal{Y})} c_{\varphi,p}^*(\nu) \Lambda_\varrho(d\nu).$$

Proposition 4.2 identifies, for each prescribed $\Lambda \in \mathcal{P}(\mathcal{P}_p(\mathcal{Y}))$, the least relative entropy $\text{KL}(\Lambda \mid \Lambda_\pi)$ among all $\varrho \in \mathcal{P}(\Theta)$ such that $\Phi_\# \varrho \equiv \Lambda_\varrho = \Lambda$, and thus (68) follows upon taking the infimum over Λ .

The Gibbs variational principle on $\mathcal{P}_p(\mathcal{Y})$, with reference measure Λ_π and cost $c_{\varphi,p}^*$, selects the unique minimizer $\Lambda_{\varphi,p}^* \ll \Lambda_\pi$, whose density satisfies the first relation in (69) (see, e.g., [11, Proposition 1.4.2]). Applying (66) to $\Lambda_{\varphi,p}^*$ then recovers the unique minimizer $\varrho_{\varphi,p}^* \ll \pi$, with density as specified in (69). $\square$

A further consequence of Corollary 4.1 is that the variational update does not distinguish between parameters inducing the same model law. Indeed, by (69), the density of $\varrho_{\varphi,p}^*$ with respect to π depends on ϑ only through $\Phi(\vartheta)$. Whenever regular conditional distributions with respect to Φ exist, the posterior minimizer therefore preserves the prior conditional distribution given Φ .

Remark 4.3. If φ is lower semicontinuous (i.e., left-continuous), then, by standard arguments, $\mathcal{R}_{\varphi,p}^*$ is lower semicontinuous with respect to weak convergence of the induced measures Λ_ϱ . Thus, the limiting incompatibility cost cannot exceed the lower limit of the approximating values.

Furthermore, linearity of integration implies that $\mathcal{R}_{\varphi,p}^*$ is affine with respect to posterior mixtures. This reflects the fact that posterior randomization is evaluated by averaging transport-based compatibility.

The reformulation (64) naturally calls for considering the second-order Wasserstein geometry of

$$\mathcal{P}_p(\mathcal{P}_p(\mathcal{Y})). \quad (70)$$

We equip this space with the p -Wasserstein distance $\mathbb{W}_p$ induced by the ground metric W_p on $\mathcal{P}_p(\mathcal{Y})$:

$$\mathbb{W}_p(\Lambda_1, \Lambda_2) := \left[\inf_{\Gamma \in \Pi(\Lambda_1, \Lambda_2)} \int_{\mathcal{P}_p(\mathcal{Y}) \times \mathcal{P}_p(\mathcal{Y})} W_p^p(\mu, \nu) \Gamma(d\mu, d\nu) \right]^{1/p}, \quad \Lambda_1, \Lambda_2 \in \mathcal{P}_p(\mathcal{P}_p(\mathcal{Y})).$$

This lifted geometry admits a natural stability estimate for asymptotic compatibility, offering a first indication of its relevance to the present variational analysis. Specifically, whenever the penalty function is Lipschitz continuous, the associated asymptotic compatibility functional inherits the same regularity with respect to the second-order p -Wasserstein metric between the induced laws. An analogous conclusion can be drawn for Hölder-continuous penalty functions.

Proposition 4.3. *Under the standing assumptions of this subsection, suppose that φ is L_φ -Lipschitz continuous for some $L_\varphi \in]0, \infty[$. Then, for every $\varrho, \tilde{\varrho} \in \mathcal{P}(\Theta)$ with $\varrho, \tilde{\varrho} \ll \pi$ and*

$$\Lambda_\varrho, \Lambda_{\tilde{\varrho}} \in \mathcal{P}_p(\mathcal{P}_p(\mathcal{Y})), \quad (71)$$

one has

$$|\mathcal{R}_{\varphi,p}^*(\varrho) - \mathcal{R}_{\varphi,p}^*(\tilde{\varrho})| \leq L_\varphi \mathbb{W}_p(\Lambda_\varrho, \Lambda_{\tilde{\varrho}}).$$

Proof. For any $\Gamma \in \Pi(\Lambda_{\varrho}, \Lambda_{\tilde{\varrho}})$, by combining (64), the L_{φ} -Lipschitz continuity of φ , the (reverse) triangle inequality for W_p , and Hölder's inequality, we deduce

$$\begin{aligned} |\mathcal{R}_{\varphi,p}^{\star}(\varrho) - \mathcal{R}_{\varphi,p}^{\star}(\tilde{\varrho})| &\leq L_{\varphi} \int_{\mathcal{P}_p(\mathcal{Y}) \times \mathcal{P}_p(\mathcal{Y})} W_p(\mu, \nu) \Gamma(d\mu, d\nu) \\ &\leq L_{\varphi} \left[\int_{\mathcal{P}_p(\mathcal{Y}) \times \mathcal{P}_p(\mathcal{Y})} W_p^p(\mu, \nu) \Gamma(d\mu, d\nu) \right]^{1/p}. \end{aligned}$$

Since Γ was arbitrary, taking the infimum over all couplings in $\Pi(\Lambda_{\varrho}, \Lambda_{\tilde{\varrho}})$ yields the desired estimate. $\square$

Remark 4.4. The p -moment requirement (71) is fairly mild. In particular, whenever it holds for the prior π , it remains valid for every posterior whose density with respect to π is essentially bounded. More generally, this property follows under suitable Hölder-type conditions on the Radon–Nikodym derivative.

A particularly transparent interpretation emerges when the minimal compatibility threshold vanishes, under a distinguished power-type specialization. In this case, the transport-induced compatibility functional is exactly the p -th power of the second-order p -Wasserstein distance between the Dirac mass concentrated at the observational law and the induced law of model distributions.

Proposition 4.4. *Under the standing assumptions of this subsection, suppose that*

$$\varepsilon_{\Theta}^{\star} = 0$$

and

$$\varphi(x) = x^p, \quad x \in \mathbb{R}_+.$$

Then, for every $\varrho \in \mathcal{P}(\Theta)$ with $\varrho \ll \pi$ and $\Lambda_{\varrho} \in \mathcal{P}_p(\mathcal{P}_p(\mathcal{Y}))$,

$$\mathcal{R}_{\varphi,p}^{\star}(\varrho) = \mathbb{W}_p^p(\delta_{\mu^{\star}}, \Lambda_{\varrho}).$$

Proof. Let $\varrho \in \mathcal{P}(\Theta)$, $\varrho \ll \pi$, satisfy $\Lambda_{\varrho} \in \mathcal{P}_p(\mathcal{P}_p(\mathcal{Y}))$. Since the only coupling between $\delta_{\mu^{\star}}$ and Λ_{ϱ} is

$$\delta_{\mu^{\star}} \otimes \Lambda_{\varrho},$$

we directly obtain

$$\mathbb{W}_p^p(\delta_{\mu^{\star}}, \Lambda_{\varrho}) = \int_{\mathcal{P}_p(\mathcal{Y})} W_p^p(\mu^{\star}, \nu) \Lambda_{\varrho}(d\nu) \equiv \mathcal{R}_{\varphi,p}^{\star}(\varrho),$$

as asserted. $\square$

Proposition 4.4 identifies the asymptotic incompatibility of a posterior distribution with the p -th power of the second-order p -Wasserstein distance between $\delta_{\mu^{\star}}$ and its transport-geometric representative. In this sense, asymptotic compatibility is governed by the geometry of the space (70).

The same geometric principle extends naturally to arbitrary compatibility thresholds, with the reference observational law replaced by a closed Wasserstein ball. With this aim, let

$$\bar{B}_{\varepsilon_{\Theta}^{\star}}(\mu^{\star}) := \{\eta \in \mathcal{P}_p(\mathcal{Y}) \mid W_p(\mu^{\star}, \eta) \leq \varepsilon_{\Theta}^{\star}\}, \quad (72)$$

and let $d_{\bar{B}_{\varepsilon_{\Theta}^{\star}}(\mu^{\star})} : \mathcal{P}_p(\mathcal{Y}) \rightarrow \mathbb{R}_+$ be the associated distance function, given by

$$d_{\bar{B}_{\varepsilon_{\Theta}^{\star}}(\mu^{\star})}(\cdot) := \inf_{\eta \in \bar{B}_{\varepsilon_{\Theta}^{\star}}(\mu^{\star})} W_p(\cdot, \eta). \quad (73)$$

Theorem 4.2. *Under the standing assumptions of this subsection, suppose that the space $(\mathcal{P}_p(\mathcal{Y}), W_p)$ is geodesic and that $\varphi(x) = x^p$, $x \in \mathbb{R}_+$. Then, for every $\varrho \in \mathcal{P}(\Theta)$ with $\varrho \ll \pi$,*

$$\mathcal{R}_{\varphi,p}^{\star}(\varrho) = \int_{\mathcal{P}_p(\mathcal{Y})} d_{\bar{B}_{\varepsilon_{\Theta}^{\star}}(\mu^{\star})}^p(\nu) \Lambda_{\varrho}(d\nu).$$

Proof. In view of (63) and (64), it suffices to prove that, for every $\nu \in \mathcal{P}_p(\mathcal{Y})$,

$$d_{\bar{B}_{\varepsilon_\Theta}^*}(\mu^*)(\nu) = [W_p(\mu^*, \nu) - \varepsilon_\Theta]_+. \quad (74)$$

The inequality “ $\geq$ ” in (74) is an immediate consequence of the metric structure of $\mathcal{P}_p(\mathcal{Y})$. Indeed, by the triangle inequality for W_p , for every $\nu \in \mathcal{P}_p(\mathcal{Y})$ and $\eta \in \bar{B}_{\varepsilon_\Theta}^*(\mu^*)$ (see (72)),

$$W_p(\nu, \eta) \geq W_p(\mu^*, \nu) - W_p(\mu^*, \eta) \geq W_p(\mu^*, \nu) - \varepsilon_\Theta,$$

whence the claim follows by (73).

The converse inequality is precisely where the assumed geodesic structure of $\mathcal{P}_p(\mathcal{Y})$ comes into play. Fix $\nu \in \mathcal{P}_p(\mathcal{Y})$. If $W_p(\mu^*, \nu) \leq \varepsilon_\Theta$, then $\nu \in \bar{B}_{\varepsilon_\Theta}^*(\mu^*)$, and both sides of (74) vanish. Otherwise, set

$$s_\nu^* := \frac{\varepsilon_\Theta}{W_p(\mu^*, \nu)} \in [0, 1[,$$

and let $(\eta_s)_{s \in [0, 1]}$ be the constant-speed Wasserstein geodesic in $\mathcal{P}_p(\mathcal{Y})$ connecting μ^* and ν . Then

$$W_p(\mu^*, \eta_{s_\nu^*}) = s_\nu^* W_p(\mu^*, \nu) \equiv \varepsilon_\Theta$$

and

$$W_p(\nu, \eta_{s_\nu^*}) = (1 - s_\nu^*) W_p(\mu^*, \nu) \equiv W_p(\mu^*, \nu) - \varepsilon_\Theta.$$

Therefore, by construction, $\eta_{s_\nu^*} \in \bar{B}_{\varepsilon_\Theta}^*(\mu^*)$ and

$$d_{\bar{B}_{\varepsilon_\Theta}^*}(\mu^*)(\nu) \leq W_p(\nu, \eta_{s_\nu^*}) = W_p(\mu^*, \nu) - \varepsilon_\Theta.$$

This completes the proof. $\square$

Remark 4.5. The geodesic assumption in Theorem 4.2 holds, for example, whenever the underlying space $(\mathcal{Y}, d_{\mathcal{Y}})$ is geodesic (as the Polish assumption is already in place); see, e.g., [27, 40, 1]. In particular, this includes the classical Euclidean setting and complete Riemannian manifolds.

The preceding results provide a natural transport-geometric characterization of asymptotic compatibility. The functional (60) factors through the push-forward map induced by Φ in (61) and thus depends on posterior candidates only through their induced distributions of model laws in Wasserstein space, while the optimal compatibility threshold determines the reference Wasserstein ball.

We conclude by illustrating these constructions through several classes of statistical models. The transport representations in the first two examples are classical; see, e.g., [40, 36].

Example 4.1 (Gaussian model laws). Let $\mathcal{Y} = \mathbb{R}^{m_{\mathcal{Y}}}$ with its Euclidean metric, and suppose that

$$\mu^* = \mathcal{N}(M^*, \Sigma^*)$$

and, for every $\vartheta \in \Theta$,

$$\nu_\vartheta = \mathcal{N}(M_\vartheta, \Sigma_\vartheta),$$

where $M_\vartheta, M^* \in \mathbb{R}^{m_{\mathcal{Y}}}$ are mean vectors and $\Sigma_\vartheta, \Sigma^* \in \mathbb{R}^{m_{\mathcal{Y}} \times m_{\mathcal{Y}}}$ are covariance matrices. Then, for $p = 2$,

$$(\mathcal{T}_\vartheta^*)^2 \equiv W_2^2(\mu^*, \nu_\vartheta) = \|M^* - M_\vartheta\|^2 + \text{tr} \left[\Sigma^* + \Sigma_\vartheta - 2 \left[(\Sigma^*)^{1/2} \Sigma_\vartheta (\Sigma^*)^{1/2} \right]^{1/2} \right]$$

(Bures–Wasserstein formula; see, e.g., [19]). Consequently, both the asymptotic compatibility profile (59) and the integrand of the associated transport-based compatibility functional (60) admit explicit expressions in terms of the corresponding mean vectors and covariance matrices.

The univariate case further elucidates the large-sample filtering mechanism developed in Section 3. Let $m_{\mathcal{Y}} = 1$, $\Theta = \mathbb{R}$, $\vartheta_0 \in \mathbb{R}$, and consider

$$\mu^* = \mathcal{N}(\vartheta_0, \sigma_0^2)$$

and, for every $\vartheta \in \mathbb{R}$,

$$\nu_{\vartheta} = \mathcal{N}(\vartheta, \sigma^2),$$

where $\sigma_0, \sigma \in]0, \infty[$, not necessarily equal. Then, still for $p = 2$,

$$W_2^2(\mu^*, \nu_{\vartheta}) = (\vartheta - \vartheta_0)^2 + (\sigma - \sigma_0)^2,$$

and hence, from (35) and (36),

$$\varepsilon_{\Theta}^* = |\sigma - \sigma_0|, \quad \Theta^* = \{\vartheta_0\}.$$

Suppose further that $\vartheta_0 \in \text{supp } \pi$. Then $\varepsilon_{\pi}^* = \varepsilon_{\Theta}^*$ (see (17) and (18)) and, for every supercritical tolerance $\tau > \varepsilon_{\Theta}^*$, the corresponding asymptotic compatibility region (24) is

$$\Theta_{\tau} = \left\{ \vartheta \in \mathbb{R} \mid |\vartheta - \vartheta_0| < \sqrt{\tau^2 - (\varepsilon_{\Theta}^*)^2} \right\}.$$

If $\sigma = \sigma_0$, then $\varepsilon_{\Theta}^* = 0$ and the model is well specified, in the sense that the limiting observational measure μ^* belongs to the model class $\{\nu_{\vartheta}\}_{\vartheta \in \mathbb{R}}$. In this case, the compatibility region reduces to

$$\Theta_{\tau} = \{ \vartheta \in \mathbb{R} \mid |\vartheta - \vartheta_0| < \tau \}.$$

If instead $\sigma \neq \sigma_0$, i.e., $\varepsilon_{\Theta}^* > 0$, the model is misspecified: no parameter is asymptotically compatible with a subcritical tolerance $\tau < \varepsilon_{\Theta}^*$, while at the critical value $\tau = \varepsilon_{\Theta}^*$ the compatibility region is empty.

Under the assumptions of Theorem 3.1, as $n \rightarrow \infty$, the finite-sample compatibility weights (21) approach the indicator $\mathbb{1}_{\Theta_{\tau}}$ of Θ_{τ} in the integrated L^1 sense (27). The associated compatibility posteriors (23) then converge in total variation, in $\mathbf{P}$ -probability, to the conditional prior distribution

$$\pi[\cdot \mid \Theta_{\tau}].$$

Thus, even though ϑ_0 is the unique asymptotically optimal parameter, a fixed supercritical tolerance alone does not in general lead to posterior concentration at ϑ_0 itself.

For a finite-sample counterpart to this behavior, fix $\vartheta_0 = 0$, $\sigma_0 = 1$, $\sigma = 1.5$, $\tau = 1$, and the prior

$$\pi = \mathcal{N}(0, 1).$$

For each $n \in \mathbb{N}^*$, given the observed sample $\mathbf{Y}^{1:n} = [\mathbf{Y}^1, \dots, \mathbf{Y}^n]$, define, on Ω ,

$$\bar{\mathbf{Y}}_n := \frac{1}{n} \sum_{i=1}^n \mathbf{Y}^i, \quad S_{\mathbf{Y},n}^2 := \frac{1}{n} \sum_{i=1}^n (\mathbf{Y}^i - \bar{\mathbf{Y}}_n)^2.$$

The corresponding probabilistic representation, as in Subsection 3.1, is

$$\nu_n(\mathbf{Y}^{1:n}) := \mathcal{N}(\bar{\mathbf{Y}}_n, S_{\mathbf{Y},n}^2).$$

For the simulated sample $\mathbf{Y}_{\vartheta}^{1:n}$, $\vartheta \in \mathbb{R}$, introduce $\bar{\mathbf{Y}}_{\vartheta,n}$ and $S_{\mathbf{Y}_{\vartheta},n}^2$ analogously, and set

$$\nu_{\vartheta,n}(\mathbf{Y}_{\vartheta}^{1:n}) := \mathcal{N}(\bar{\mathbf{Y}}_{\vartheta,n}, S_{\mathbf{Y}_{\vartheta},n}^2).$$

For every $\vartheta \in \mathbb{R}$, the resulting squared finite-sample discrepancy takes the form

$$W_2^2(\nu_n(\mathbf{Y}^{1:n}), \nu_{\vartheta,n}(\mathbf{Y}_{\vartheta}^{1:n})) = (\bar{\mathbf{Y}}_n - \bar{\mathbf{Y}}_{\vartheta,n})^2 + (S_{\mathbf{Y},n} - S_{\mathbf{Y}_{\vartheta},n})^2,$$

with $S_{\mathbf{Y},n}$ and $S_{\mathbf{Y}_{\vartheta},n}$ denoting the respective empirical standard deviations. In particular,

$$\varepsilon_{\Theta}^* = 1/2, \quad \Theta_{\tau} =]-\sqrt{3}/2, \sqrt{3}/2[.$$

For each $n \in \{50, 200, 1000\}$, the observed sample consists of the first n observations from a sequence of independent $\mathcal{N}(0, 1)$ random variables, generated with random seed 20261010. The compatibility weights are estimated from 10^5 independent Monte Carlo replicates of Gaussian sample standard deviations,

exploiting their independence from the sample means to integrate analytically over the latter. The posterior densities are obtained by normalizing these weights with respect to the prior.

Figure 1 presents the convergence of finite-sample compatibility weights and posterior densities toward their limiting profiles, together with the attenuation of finite-sample asymmetries.

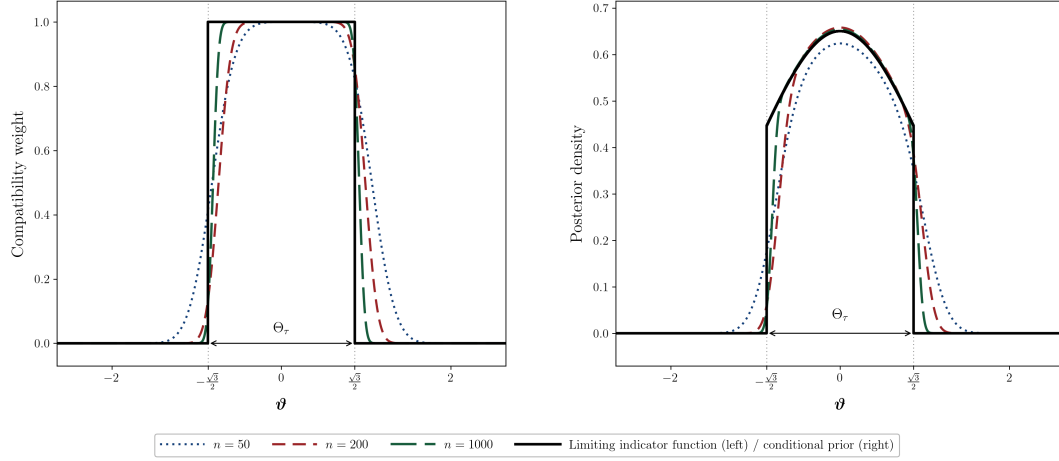


Figure 1: Finite-sample compatibility filtering in the misspecified Gaussian model for $n = 50, 200, 1000$.

Example 4.2 (Finite-state models). Consider the case where $\mathcal{Y} = \{y_1, \dots, y_m\} \subset \mathbb{R}^{m_y}$, $m \in \mathbb{N}^*$, and let

$$\mu^\star = \sum_{i=1}^m q_i^\star \delta_{y_i}$$

and, for every $\vartheta \in \Theta$,

$$\nu_\vartheta = \sum_{i=1}^m q_{\vartheta,i} \delta_{y_i},$$

where $q_i^\star$ and $q_{\vartheta,i}$, $i = 1, \dots, m$, denote the corresponding probability weights. Then

$$W_p^p(\mu^\star, \nu_\vartheta) = \min_{\substack{\gamma_{ij} \geq 0 \\ \sum_{j=1}^m \gamma_{ij} = q_i^\star \\ \sum_{i=1}^m \gamma_{ij} = q_{\vartheta,j}}} \sum_{i,j=1}^m \gamma_{ij} d_y^p(y_i, y_j).$$

Hence, the asymptotic compatibility profile is determined by finite-dimensional optimal transport problems, while the associated functional is obtained by integrating the corresponding transformed costs with respect to the parameter distribution. This covers, among others, categorical and finite-state models.

The formulation illustrated by Examples 4.1 and 4.2 also applies to mixed discrete–continuous laws. For instance, a zero-inflated model may take the form

$$\nu_\vartheta = a_\vartheta \delta_{y_0} + (1 - a_\vartheta) \eta_\vartheta, \quad a_\vartheta \in [0, 1],$$

where $y_0 \in \mathcal{Y}$, and $\eta_\vartheta \in \mathcal{P}_p(\mathcal{Y})$ may be, for example, a Gaussian distribution or a finite Gaussian mixture. The corresponding Wasserstein discrepancy is defined directly between the resulting hybrid probability measures, without requiring separate transport criteria for their discrete and continuous components.

Deterministic models are recovered as an immediate special case. Indeed, if

$$\mu^\star = \delta_{y^\star}$$

and, for every $\vartheta \in \Theta$,

$$\nu_\vartheta = \delta_{y_\vartheta},$$

where $y^\star, y_\vartheta \in \mathcal{Y}$, then

$$W_p(\mu^\star, \nu_\vartheta) = d_{\mathcal{Y}}(y^\star, y_\vartheta).$$

Therefore, in the deterministic setting, the Wasserstein discrepancy reduces exactly to the underlying distance between the observational and model outputs.

5 Conclusions

In this paper, we introduced a measure-theoretic formulation of discrepancy-based Bayesian inference through nonnegative compatibility weights. This approach provides a common basis for studying posterior asymptotics and the associated variational and geometric structures.

The analysis distinguishes two asymptotic regimes. For vanishing thresholds, convergence of suitably normalized weights yields a total variation limit for the corresponding posterior distributions. In the large-sample regime, a fixed tolerance above the critical threshold leads, under appropriate assumptions, to the prior conditioned on an asymptotic compatibility region. Complementary persistence bounds apply under weaker conditions. These results show that increasing the sample size does not necessarily imply posterior concentration. In particular, a positive tolerance may preserve a nontrivial distribution over compatible parameter values, even when the limiting discrepancy has a unique minimizer.

The variational analysis connects compatibility-based inference with entropy-regularized optimization. The classical Gibbs principle characterizes posterior distributions as minimizers of compatibility-based losses penalized by relative entropy. We also establish stability of these minimization problems as the threshold vanishes. For Wasserstein discrepancies, a distinguished class of asymptotic compatibility costs admits a geometric representation through distances from a Wasserstein ball. The variational problem admits an exact reduction to probability distributions over model laws, preserving the minimum value and the correspondence between minimizers. The resulting functionals can be interpreted as expected incompatibility costs, without requiring the axioms of monetary risk measures.

Several directions remain open. These include joint asymptotic regimes involving different rates of tolerance decay as the sample size increases, finer results on posterior concentration and variational convergence, and extensions to more general discrepancies and transport costs. Further developments may concern conditional or sequential models and settings in which the limiting observational law is uncertain. The geometric formulation also suggests connections with distributionally robust optimization and multivariate risk assessment; see, e.g., [29, 18, 25].

References

- [1] Agueh, M., & Carlier, G. (2011). *Barycenters in the Wasserstein space*. SIAM Journal on Mathematical Analysis, 43(2), 904–924.
- [2] Ambrosio, L., Gigli, N., & Savaré, G. (2008). *Gradient Flows: In Metric Spaces and in the Space of Probability Measures*. Birkhäuser, Basel.
- [3] Beaumont, M. A., Zhang, W., & Balding, D. J. (2002). *Approximate Bayesian computation in population genetics*. Genetics, 162, 2025–2035.
- [4] Bernton, E., Jacob, P. E., Gerber, M., & Robert, C. P. (2019). *Approximate Bayesian computation with the Wasserstein distance*. Journal of the Royal Statistical Society: Series B (Statistical Methodology), 81(2), 235–269.
- [5] Bissiri, P. G., Holmes, C. C., & Walker, S. G. (2016). *A General Framework for Updating Belief Distributions*. Journal of the Royal Statistical Society: Series B (Statistical Methodology), 78(5), 1103–1130.
- [6] Boissard, E., & Le Gouic, T. (2014). *On the mean speed of convergence of empirical and occupation measures in Wasserstein distance*. Annales de l’Institut Henri Poincaré, Probabilités et Statistiques, 50(2), 539–563.

- [7] Chen, S., Mira, A., & Onnela, J.-P. (2020). *Flexible model selection for mechanistic network models*. Journal of Complex Networks, 8(2), cnz024.
- [8] Dal Maso, G. (1993). *An Introduction to Γ -Convergence*. Progress in Nonlinear Differential Equations and Their Applications, Vol. 8. Birkhäuser, Boston.
- [9] Dembo, A., & Zeitouni, O. (1998). *Large Deviations Techniques and Applications*. 2nd ed. Applications of Mathematics, Vol. 38. Springer, New York.
- [10] Donsker, M. D., & Varadhan, S. R. S. (1975). *Asymptotic evaluation of certain Markov process expectations for large time, I*. Communications on Pure and Applied Mathematics, 28(1), 1–47.
- [11] Dupuis, P., & Ellis, R. S. (1997). *A Weak Convergence Approach to the Theory of Large Deviations*. Wiley Series in Probability and Statistics. John Wiley & Sons, New York.
- [12] Dutta, R., Mira, A., & Onnela, J.-P. (2018). *Bayesian inference of spreading processes on networks*. Proceedings of the Royal Society A: Mathematical, Physical and Engineering Sciences, 474(2215), 20180129.
- [13] Ebert, A., Dutta, R., Mengersen, K., Mira, A., Ruggeri, F., & Wu, P. (2018). *Likelihood-free parameter estimation for dynamic queueing networks: Case study of passenger flow in an international airport terminal*. arXiv preprint, arXiv:1804.06374.
- [14] Fearnhead, P., & Prangle, D. (2012). *Constructing summary statistics for approximate Bayesian computation: semi-automatic approximate Bayesian computation*. Journal of the Royal Statistical Society: Series B (Statistical Methodology), 74(3), 419–474.
- [15] Fournier, N., & Guillin, A. (2015). *On the rate of convergence in Wasserstein distance of the empirical measure*. Probability Theory and Related Fields, 162(3–4), 707–738.
- [16] Frazier, D. T., Martin, G. M., Robert, C. P., & Rousseau, J. (2018). *Asymptotic properties of approximate Bayesian computation*. Biometrika, 105(3), 593–607.
- [17] Frazier, D. T., Robert, C. P., & Rousseau, J. (2020). *Model misspecification in approximate Bayesian computation: Consequences and diagnostics*. Journal of the Royal Statistical Society: Series B (Statistical Methodology), 82(2), 421–444.
- [18] Gao, R., & Kleywegt, A. J. (2023). *Distributionally robust stochastic optimization with Wasserstein distance*. Mathematics of Operations Research, 48(2), 603–655.
- [19] Gelbrich, M. (1990). *On a formula for the L^2 Wasserstein metric between measures on Euclidean and Hilbert spaces*. Mathematische Nachrichten, 147(1), 185–203.
- [20] Goyal, R., & Onnela, J.-P. (2023). *Framework for converting mechanistic network models to probabilistic models*. Journal of Complex Networks, 11(5), cnad034.
- [21] Jiang, B., Wu, T.-Y., & Wong, W. H. (2018). *Approximate Bayesian computation with Kullback–Leibler divergence as data discrepancy*. Proceedings of Machine Learning Research, 84, 1711–1721.
- [22] Kantorovich, L. V. (1942). *On the translocation of masses*. Doklady Akademii Nauk SSSR, 37(7–8), 227–229.
- [23] Lin, T., Ho, N., Chen, X., Cuturi, M., & Jordan, M. I. (2020). *Revisiting fixed-support Wasserstein barycenters: Computational hardness and efficient algorithms*. Advances in Neural Information Processing Systems, 33, 5368–5380.
- [24] Marin, J.-M., Pudlo, P., Robert, C. P., & Ryder, R. J. (2012). *Approximate Bayesian computational methods*. Statistics and Computing, 22, 1167–1180.

- [25] Mastrogiacono, E., Rocca, M., & Tarsia, M. (2026). *Multi-Objective Optimization and its Connection to Multivariate Risk Measures*. Journal of Optimization Theory and Applications, 209(1), 30.
- [26] Matsubara, T., Knoblauch, J., Briol, F.-X., & Oates, C. J. (2022). *Robust generalised Bayesian inference for intractable likelihoods*. Journal of the Royal Statistical Society: Series B (Statistical Methodology), 84(3), 997–1022.
- [27] McCann, R. J. (1997). *A convexity principle for interacting gases*. Advances in Mathematics, 128(1), 153–179.
- [28] Miller, J. W., & Dunson, D. B. (2019). *Robust Bayesian inference via coarsening*. Journal of the American Statistical Association, 114(527), 1113–1125.
- [29] Mohajerin Esfahani, P., & Kuhn, D. (2018). *Data-driven distributionally robust optimization using the Wasserstein metric: Performance guarantees and tractable reformulations*. Mathematical Programming, 171(1–2), 115–166.
- [30] Nguyen, H. D., Arbel, J., Lü, H., & Forbes, F. (2019). *Approximate Bayesian computation via the energy statistic*. IEEE Transactions on Signal Processing, 67(19), 5005–5019.
- [31] Pacchiardi, L., Khoo, S., & Dutta, R. (2024). *Generalized Bayesian likelihood-free inference*. Electronic Journal of Statistics, 18(2), 3628–3686.
- [32] Panaretos, V. M., & Zemel, Y. (2019). *Statistical aspects of Wasserstein distances*. Annual Review of Statistics and Its Application, 6, 405–431.
- [33] Parthasarathy, K. R. (1967). *Probability Measures on Metric Spaces*. Academic Press, New York–London.
- [34] Park, M., Jitkrittum, W., & Sejdinovic, D. (2016). *K2-ABC: Approximate Bayesian computation with kernel embeddings*. Proceedings of Machine Learning Research, 51, 398–407.
- [35] Peyré, G., & Cuturi, M. (2019). *Computational Optimal Transport*. Foundations and Trends in Machine Learning, 11(5–6), 355–607.
- [36] Santambrogio, F. (2015). *Optimal Transport for Applied Mathematicians: Calculus of Variations, PDEs, and Modeling*. Birkhäuser, Cham.
- [37] Sisson, S. A., Fan, Y., & Beaumont, M. A. (2019). *Handbook of Approximate Bayesian Computation*. Chapman & Hall/CRC, Boca Raton.
- [38] Sriperumbudur, B. K., Gretton, A., Fukumizu, K., Schölkopf, B., & Lanckriet, G. R. G. (2010). *Hilbert space embeddings and metrics on probability measures*. Journal of Machine Learning Research, 11, 1517–1561.
- [39] Villani, C. (2003). *Topics in Optimal Transportation*. American Mathematical Society, Providence, RI.
- [40] Villani, C. (2009). *Optimal Transport: Old and New*. Springer, Berlin.
- [41] Weed, J., & Bach, F. (2019). *Sharp asymptotic and finite-sample rates of convergence of empirical measures in Wasserstein distance*. Bernoulli, 25(4A), 2620–2648.
- [42] Wilkinson, R. D. (2013). *Approximate Bayesian computation (ABC) gives exact results under the assumption of model error*. Statistical Applications in Genetics and Molecular Biology, 12(2), 129–141.